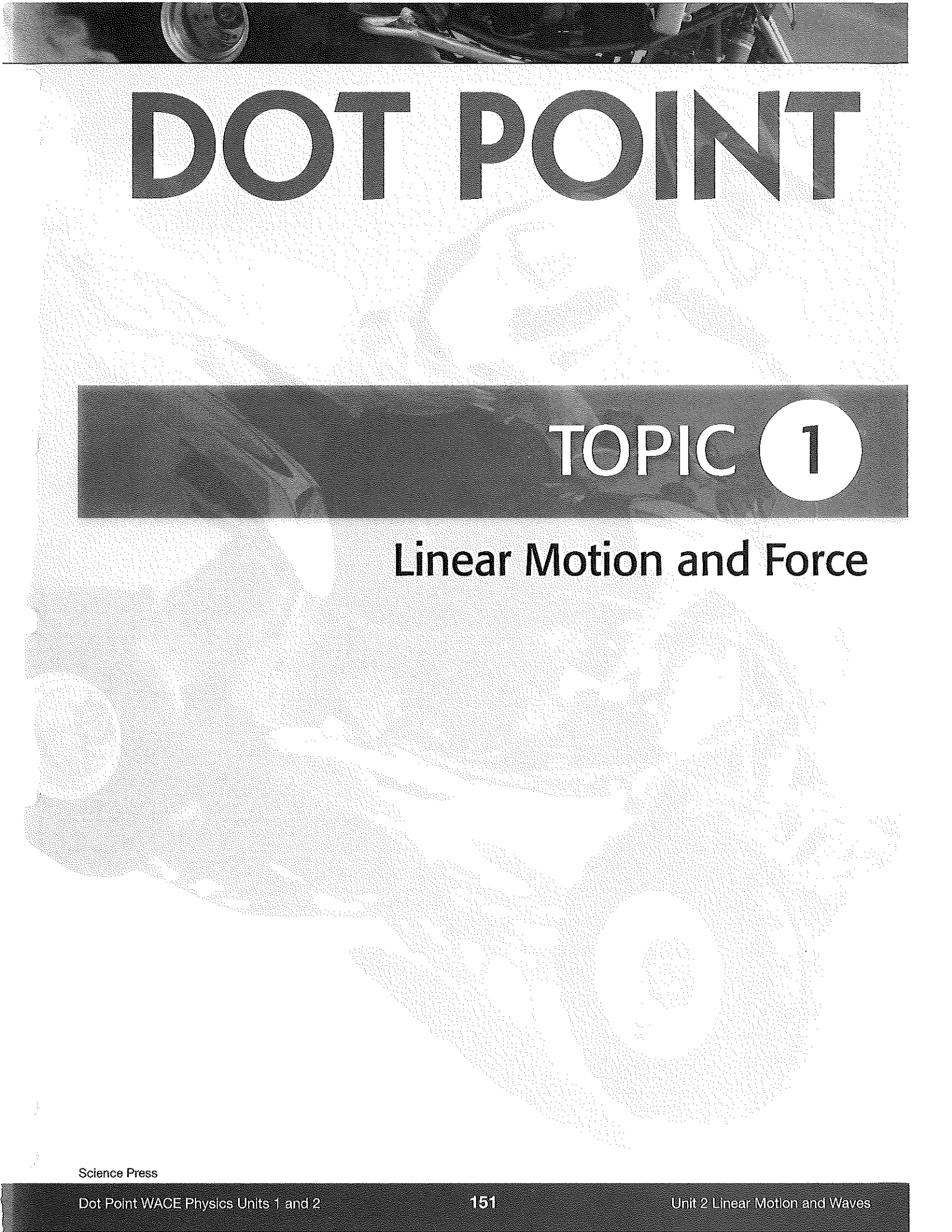




DOT POINT

TOPIC 1

Linear Motion and Force

1.1 Uniformly accelerated motion and quantities.

1.1.1 Understanding the quantities.

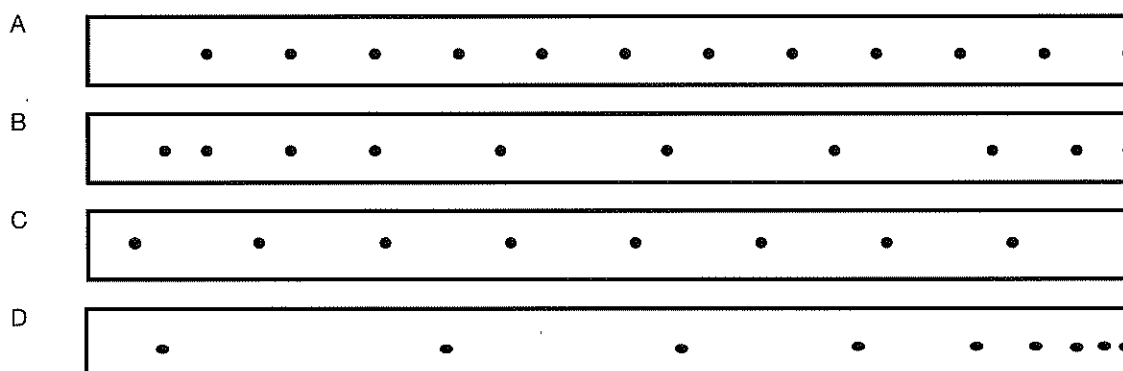
1.1.1.1 Define each of the following quantities and identify each as a scalar or vector quantity.

Quantity	Definition	Scalar or vector
Distance travelled		
Displacement		
Speed		
Velocity		
Acceleration		

1.1.1.2 What do we mean by 'uniform' motion or 'uniform' acceleration?

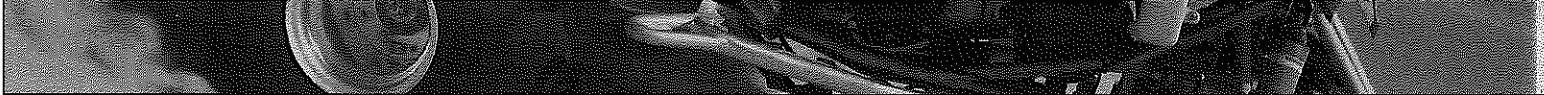
1.1.1.3 Explain, giving an appropriate example, how the velocity of an object moving at 30 m s^{-1} can change and yet it still continues to move at 30 m s^{-1} .

1.1.1.4 The ticker timer tapes show the motion of four objects starting from the right and moving to the left.



(a) Which tapes indicate the objects moving with constant speed?

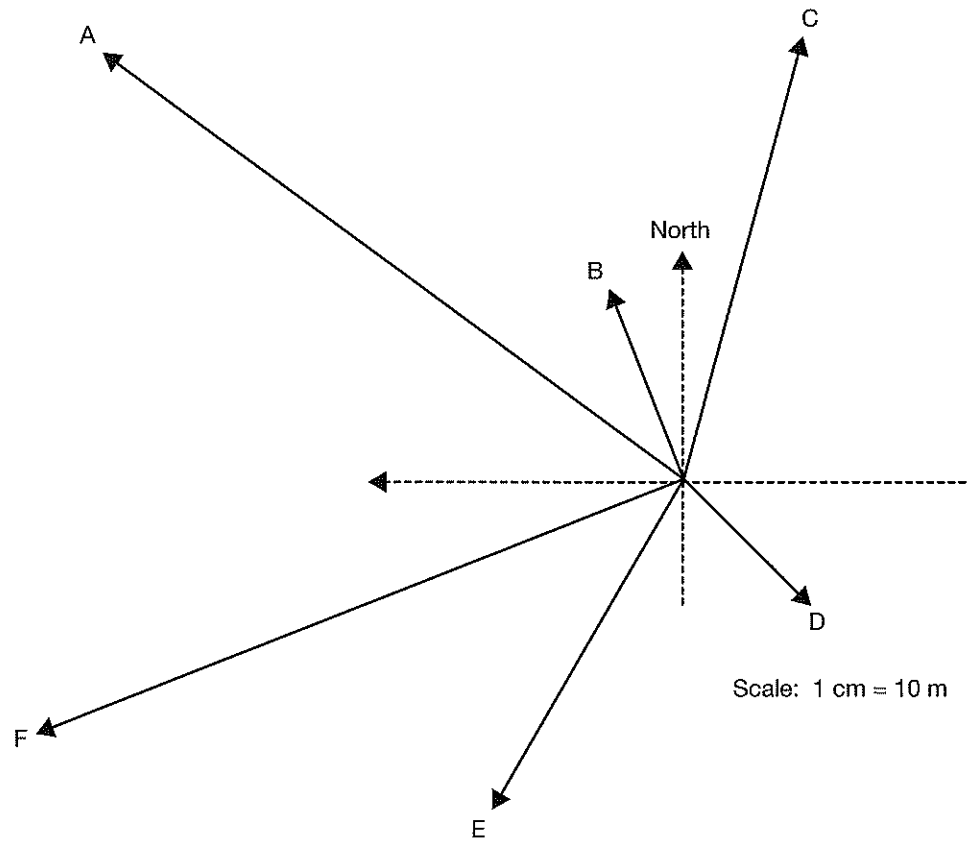
(b) In your own words, describe the motion of the objects indicated by the other tapes.



1.1.1.5 Distinguish between the different speeds below.

Constant velocity	
Average velocity	
Instantaneous velocity	
Initial velocity	
Final velocity	

1.1.1.6 Expressing the directions as bearings, calculate the displacement represented by each of the vectors in the diagram. Each snail started at the origin. (Don't forget the scale!)



Vector	Displacement (m)	Direction as compass direction	Direction as bearing
A			
B			
C			
D			
E			
F			

1.1.1.7 Describe two speed changes which occur during typical journeys in a car and state one reason for each change.

1.1.1.8 Compare average and instantaneous speed.

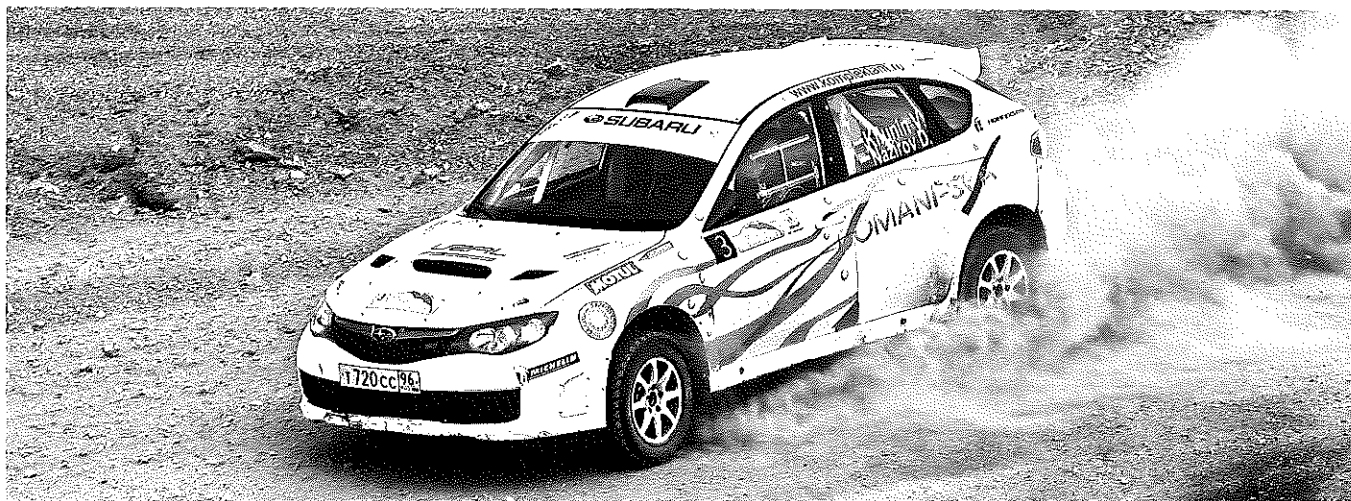
1.1.1.9 Compare scalar and vector quantities.

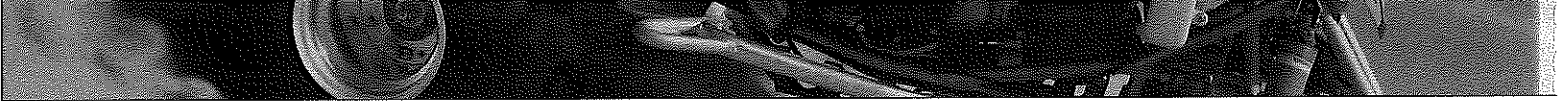
1.1.1.10 Complete the table to show five examples each of scalar and vector quantities.

Scalar quantities	Symbol	SI unit	Vector quantities	Symbol	SI unit

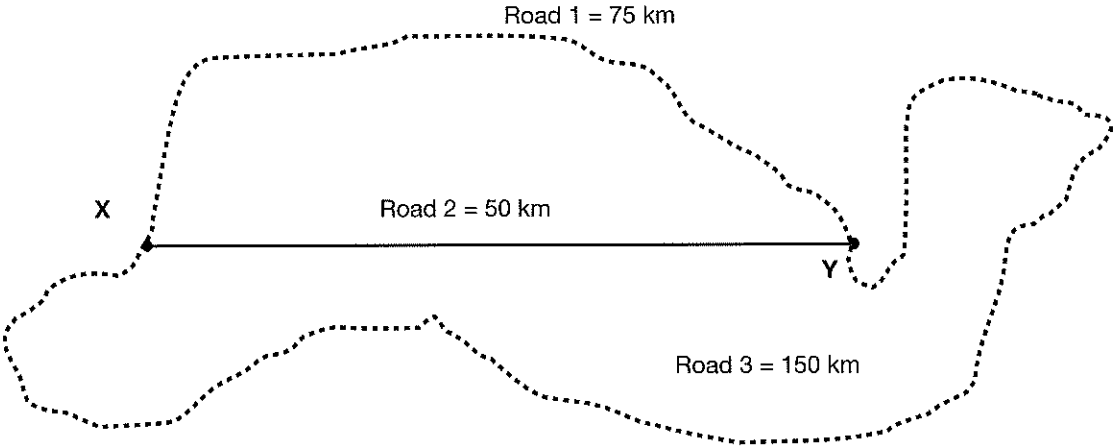
1.1.1.11 Compare the distance travelled by an object with its displacement.

1.1.1.12 Clarify the idea of expressing direction of travel as a bearing.





1.1.1.13 Three objects travel from X to Y by three different roads as shown in the diagram. Y is due east of X.

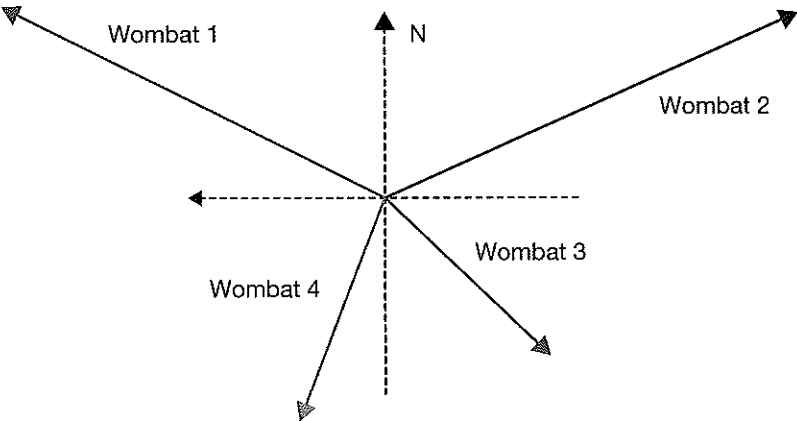


Complete the table to show information about the three trips.

Object travelling by	Distance travelled (km)	Displacement (km)
Road 1		
Road 2		
Road 3		

1.1.1.14 The diagram shows the paths taken by four wombats as they came out of their burrow to search for food. The diagram is drawn to scale where 1 cm = 10 m.

Complete the table to show information about the four wombats' trips.



Wombat	Distance travelled (m)	Displacement (m) (directions as compass readings)	Displacement (m) (directions as bearings)
1			
2			
3			
4			

1.1.2 The use of SI units to quantify descriptions of motion.

1.1.2.1 What are SI units and why are they used?

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1.1.2.2 List the rules we should use when using SI units.

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1.1.2.3 Units like kilometres per hour, or light years, or hours are not SI units, but we often use them in physics. Explain why.

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1.1.2.4 Complete the table to give the SI units most commonly used in calculations for the quantities listed.

Quantity	SI unit (name and symbol)	Quantity	SI unit (name and symbol)
Mass		Electric potential difference	
Length		Electrical resistance	
Time		Electric current	
Displacement		Speed	
Force		Velocity	
Energy		Acceleration	
Power		Temperature	
Momentum		Volume	

1.1.2.5 Complete the table.

Unit prefix	Symbol	Meaning	Unit prefix	Symbol	Meaning
deca					10^{-1}
hecta					10^{-2}
	k			m	
	M			μ	
		10^9	nano		
		10^{12}	pico		
		10^{15}	femto		
	E			a	
zeta					10^{-21}
yotta					10^{-24}

1.2 Graphs, vectors and equations of motion.

1.2.1 Equations of motion and vector quantities.

1.2.1.1 Recall the equations we use for each of the following quantities (if the quantity is listed more than once, then different equations are required for each listing).

- (a) Average speed
- (b) Average speed
- (c) Average velocity
- (d) Average velocity
- (e) Acceleration
- (f) Acceleration
- (g) Final velocity

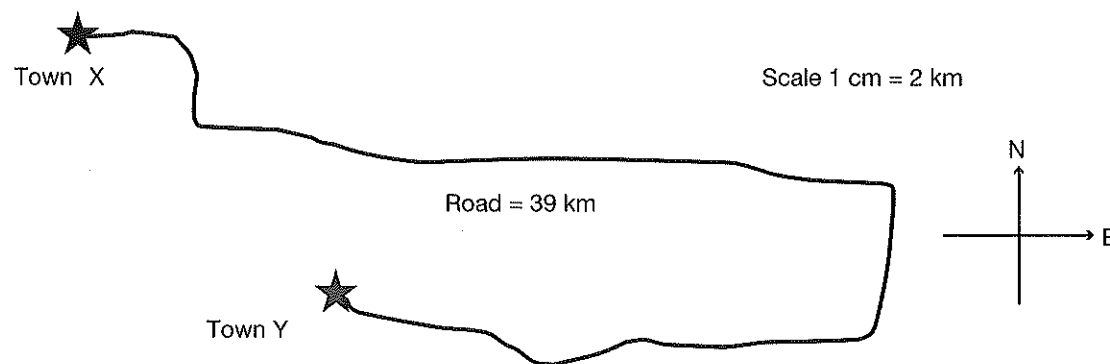
1.2.1.2 A swimmer, travelling at a steady rate, swims 100 m in 59 s. Calculate her average speed.

1.2.1.3 A rocket rises 2500 m in 11.2 s. Calculate its average speed.

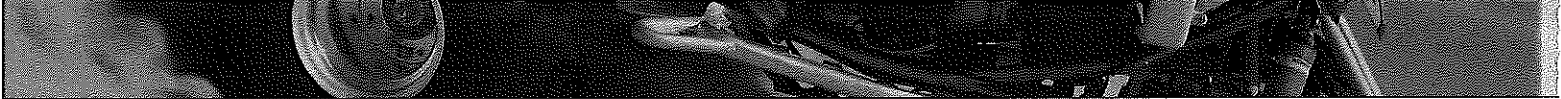
1.2.1.4 A car travelled 800 km east in 16.5 hours. Calculate its average speed and velocity.

1.2.1.5 A ball rolled 2.4 m from X to Y. This took 4.2 s. Calculate its average speed and velocity.

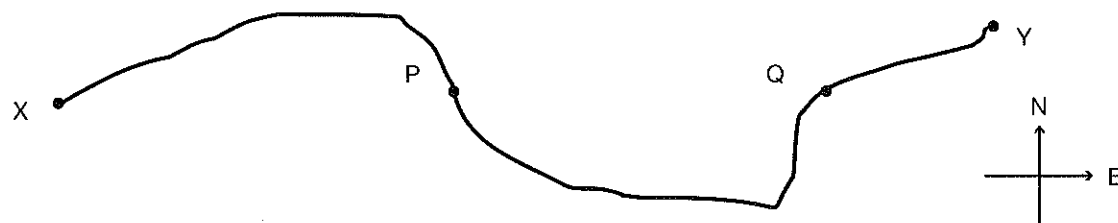
1.2.1.6 A cyclist left P and rode 39 km along the road connecting town X to town Y. Because of bends and a river, the road was longer than the actual distance between X and Y which was only 8 km. The journey took 90 minutes.



- (a) How far does the cyclist travel?
- (b) What is her displacement when she is at Y?
- (c) Calculate the cyclist's average speed for the trip.
- (d) Calculate her average velocity.

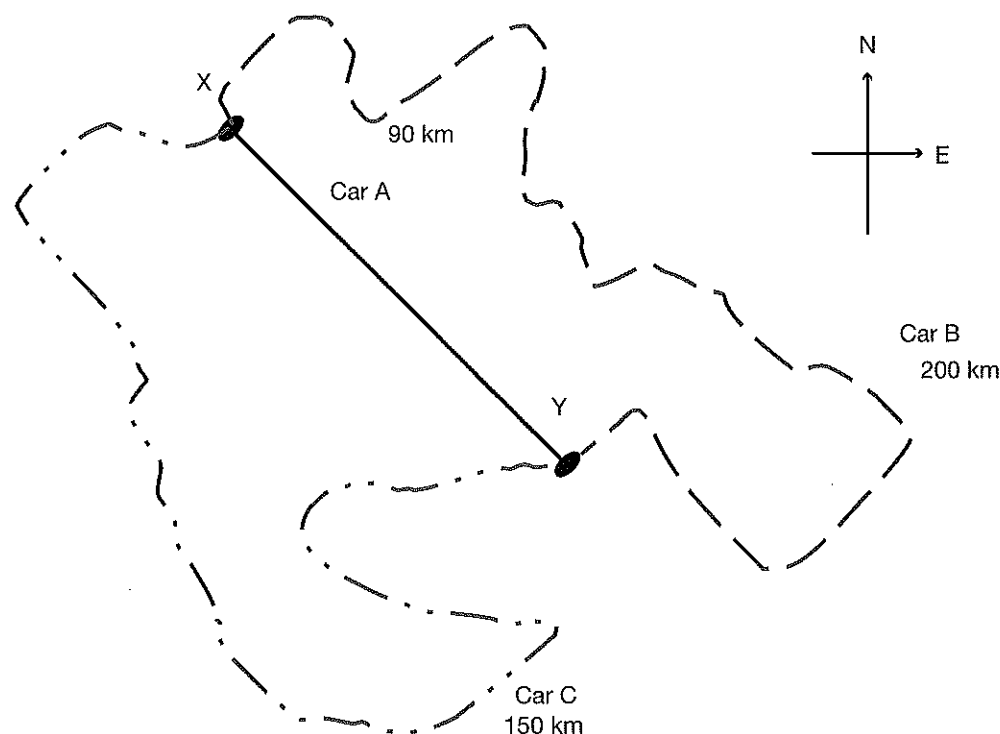


- 1.2.1.7** A car takes 3 hours to travel along the 180 km road from X and Y as shown in the diagram at a constant speed of 60 km h^{-1} . X and Y are 120 km apart.



- (a) Compare the car's instantaneous speed at points P and Q.
- (b) Compare the car's instantaneous velocity at points P and Q.
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- (c) Calculate the car's average speed for the journey.
- (d) Calculate the car's average velocity for the journey.

- 1.2.1.8** Consider three cars which started at town X and travelled to town Y by three different roads as shown in the diagram. Car A travelled directly from X to Y, a distance of 90 km in 2 hours. Car B made its 200 km trip in 3 hours, while car C took 5 hours to travel the 150 km from X to Y by the route it took.



- (a) How far did car A travel?
- (b) How long did this take?
- (c) On average, how far did car A travel each hour?

- (d) Calculate the average speed of car A.
- (e) Calculate the average speed of car B.
- (f) Calculate the average speed of car C.
- (g) Explain why we are only talking about *average* speeds here.

- (h) Calculate the average *velocity* of each car for the journey from X to Y.

A =

B =

C =

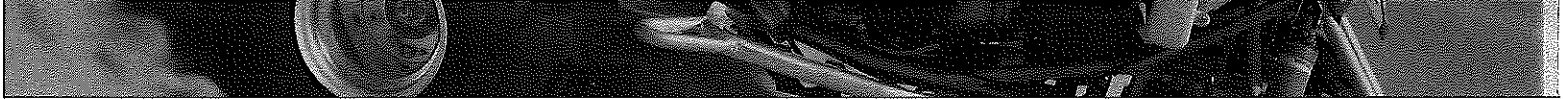
1.2.1.9 A racing car is attempting to break the 'standing kilometre' time record. When the starting light turns green, it accelerates at maximum rate and crosses the finish line at 180 km h^{-1} 40 s later (on a racing track of course).

- (a) Identify the initial speed of the car.
- (b) Identify its final speed.
- (c) Calculate its average speed.
- (d) Estimate its instantaneous speed 20 s after starting. Justify your answer.

- (e) Predict the constant speed to cover the same distance in the same time.

1.2.1.10 Complete the table.

	Displacement	Time taken	Average speed	Average velocity
(a)	150 m north	4.0 s		
(b)	20 m south	0.25 s		
(c)	96 m east	3 s		
(d)	2.8 m SE	0.5 s		
(e)	36 m bearing 147°	1.5 s		
(f)	600 m bearing 112°		50 m s^{-1}	
(g)	80 m bearing 347°		15 m s^{-1}	
(h)	25 km bearing 220°		5 km h^{-1}	
(i)	2.6 km NE		80 m s^{-1}	
(j)	1.5 km south		15 km h^{-1}	
(k)		1.75 hr		$60 \text{ km h}^{-1} \text{ E}$
(l)		18 s		$7.5 \text{ m s}^{-1} \text{ NE}$
(m)		5 min		$9 \text{ m s}^{-1} \text{ SW}$
(n)		3.5 s		$0.3 \text{ m s}^{-1} \text{ W}$
(o)		0.2 s		$28 \text{ m s}^{-1} \text{ bearing } 045^\circ$



1.2.1.11 Outline the conditions under which the equations for uniformly accelerated motion may be applied.

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1.2.1.12

(a) Sometimes we talk about 'positive' and 'negative' acceleration. Explain what is meant by these terms.

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(b) Give an example of an object that has a negative velocity and a positive acceleration.

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(c) Give an example of an object that has a positive velocity and a negative acceleration.

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1.2.1.13 State the average value of the acceleration due to gravity near the Earth's surface.

1.2.1.14

(a) An object falls from a height of 30 000 m towards the surface of Earth. Is it correct to say that its acceleration is 9.8 m s^{-2} downwards? Explain your answer.

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(b) There is another factor which affects the rate of fall of an object apart from the value of the acceleration due to gravity. Identify this factor and state its effect.

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1.2.1.15 A car, at rest, accelerates at 6 m s^{-2} N for 15 s. Calculate its velocity after:

(a) 3 seconds

(b) 6 seconds

(c) 9 seconds

1.2.1.16 A car moving at 5.75 m s^{-1} W hits a wall and stops in 0.015 s. Calculate the acceleration stopping the car.

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1.2.1.17 A rocket accelerates at 22 m s^{-2} until its speed is 858 m s^{-1} . How long does this take?

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1.2.1.18 After 6 s of accelerating at 3.0 m s^{-2} , a car moves at 46 m s^{-1} . Calculate its initial speed if the acceleration was:

- (a) Positive.
- (b) Negative.

1.2.1.19 A car accelerates at $3.5 \text{ m s}^{-2} \text{ S}$ for 9 s. After this time it is moving at $25 \text{ m s}^{-1} \text{ S}$. Calculate its initial velocity.

1.2.1.20 A car accelerates at $3 \text{ m s}^{-2} \text{ N}$ for 45 s. After this time the car is moving at $150 \text{ m s}^{-1} \text{ S}$. Calculate its initial velocity.

1.2.1.21 A car is dropped from rest from a height of 75 m in an experiment to examine its safety features. Determine:

- (a) How long it takes to hit the ground.
- (b) Its speed just before it hits the ground.
- (c) Its average speed while falling.

1.2.1.22 A crash dummy is dropped from rest from a hot air balloon in an experiment to examine what would happen to a person crashing into a solid wall at 30 km h^{-1} .

- (a) Express 30 km h^{-1} in m s^{-1}
- (b) Find the height from which the dummy was dropped.

1.2.1.23 A ball is dropped from rest from a hot air balloon. It hits the ground 15 s later.

- (a) Calculate the speed with which it hits the ground.
- (b) Find the height from which the ball was dropped.



1.2.2 Representation of vector and scalar quantities.

1.2.2.1 Distinguish between a scalar and a vector quantity.

1.2.2.2 Complete the table to list scalar and vector quantities and the main units we use to measure them.

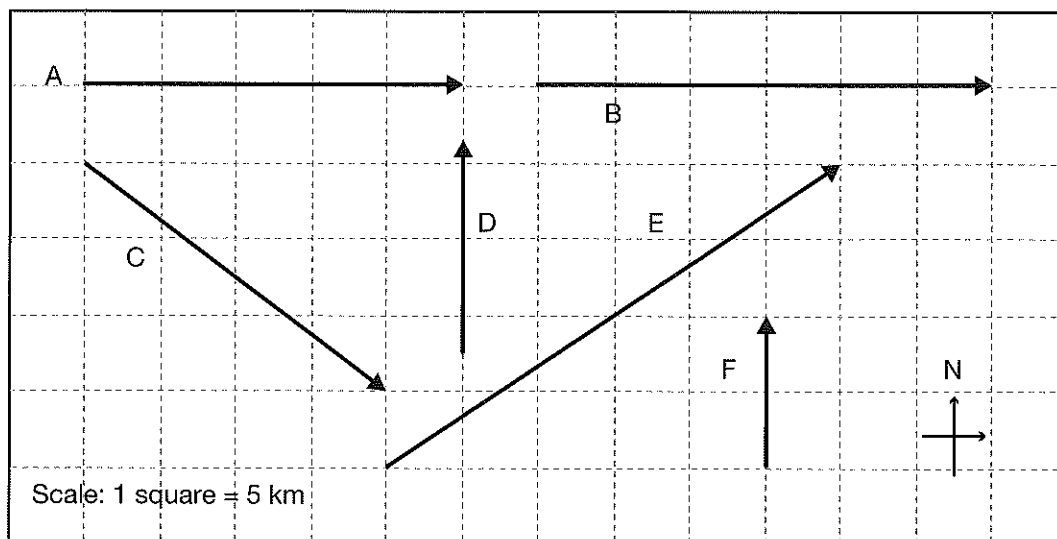
Scalar quantities			Vector quantities		
Quantity	Symbol	Unit	Quantity	Symbol	Unit

1.2.2.3 Vector X is 12 N right, vector Y is 8 N left, and vector Z is 17 N left. What is:

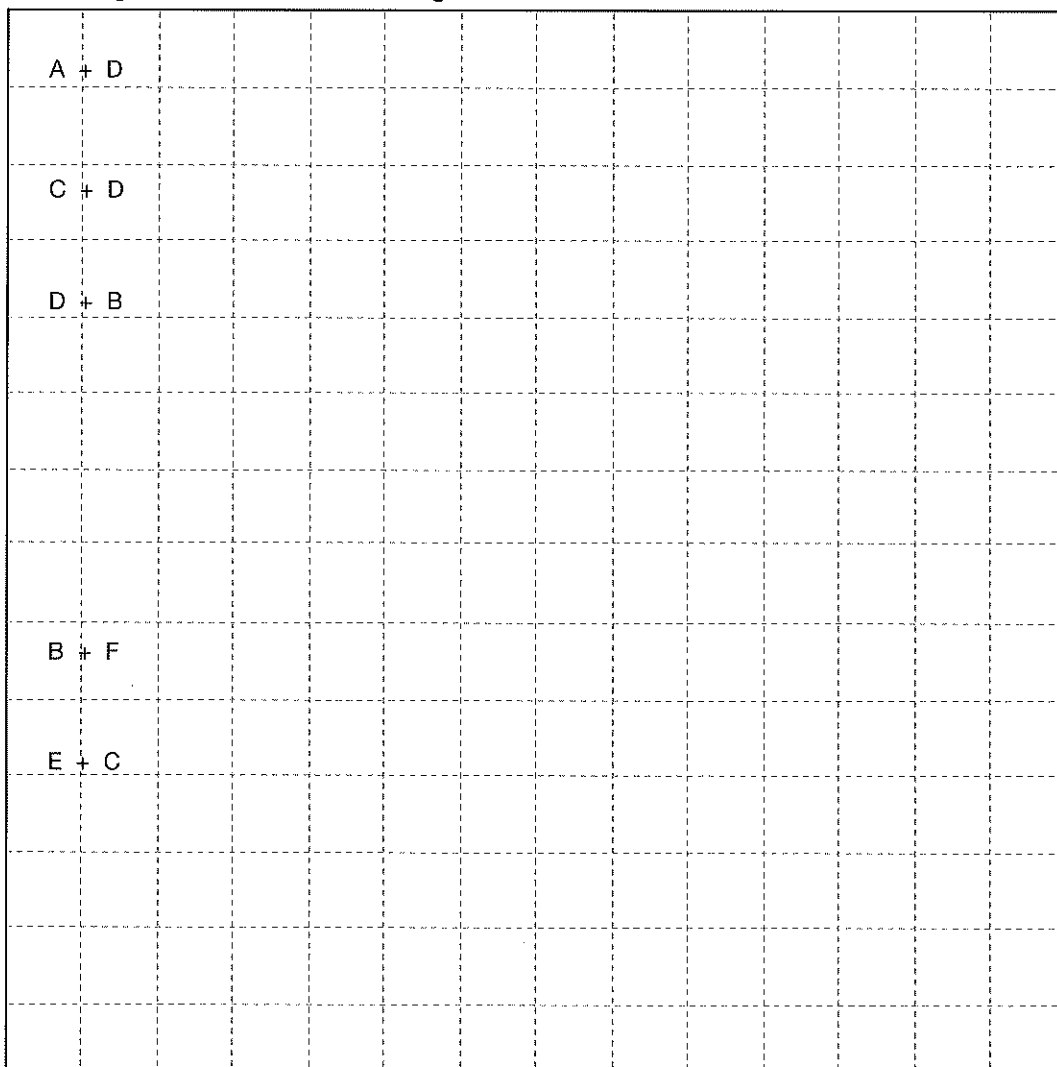
- (a) $X + Y + Z$?
- (b) $X - Y - Z$?
- (c) $Y - Z + X$?

1.2.2.4 A vector of magnitude 6 units is added to a second vector of magnitude 5 units. What are the maximum and minimum possible values for this addition? Justify your answer.

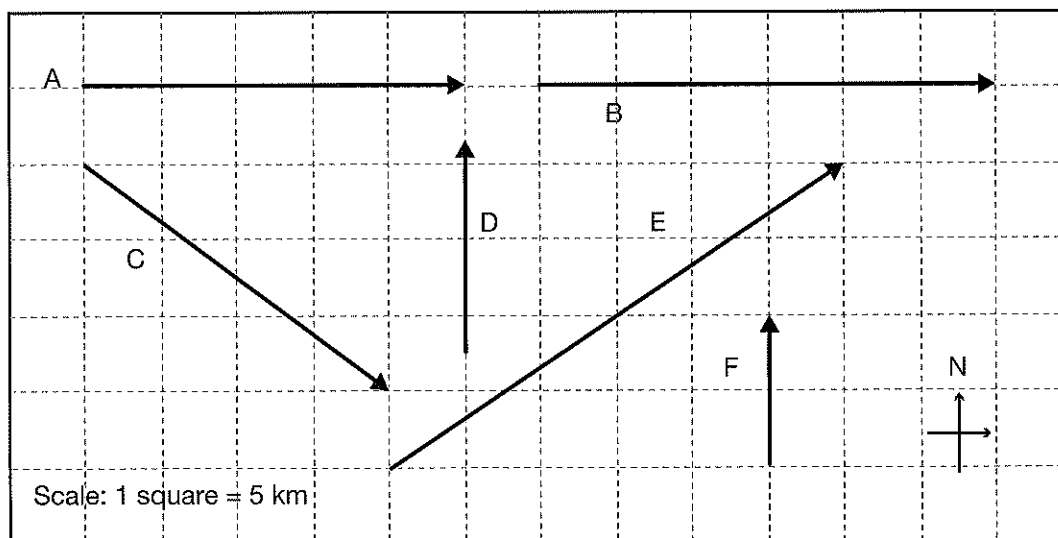
1.2.2.5 The diagram represents several displacement vectors.



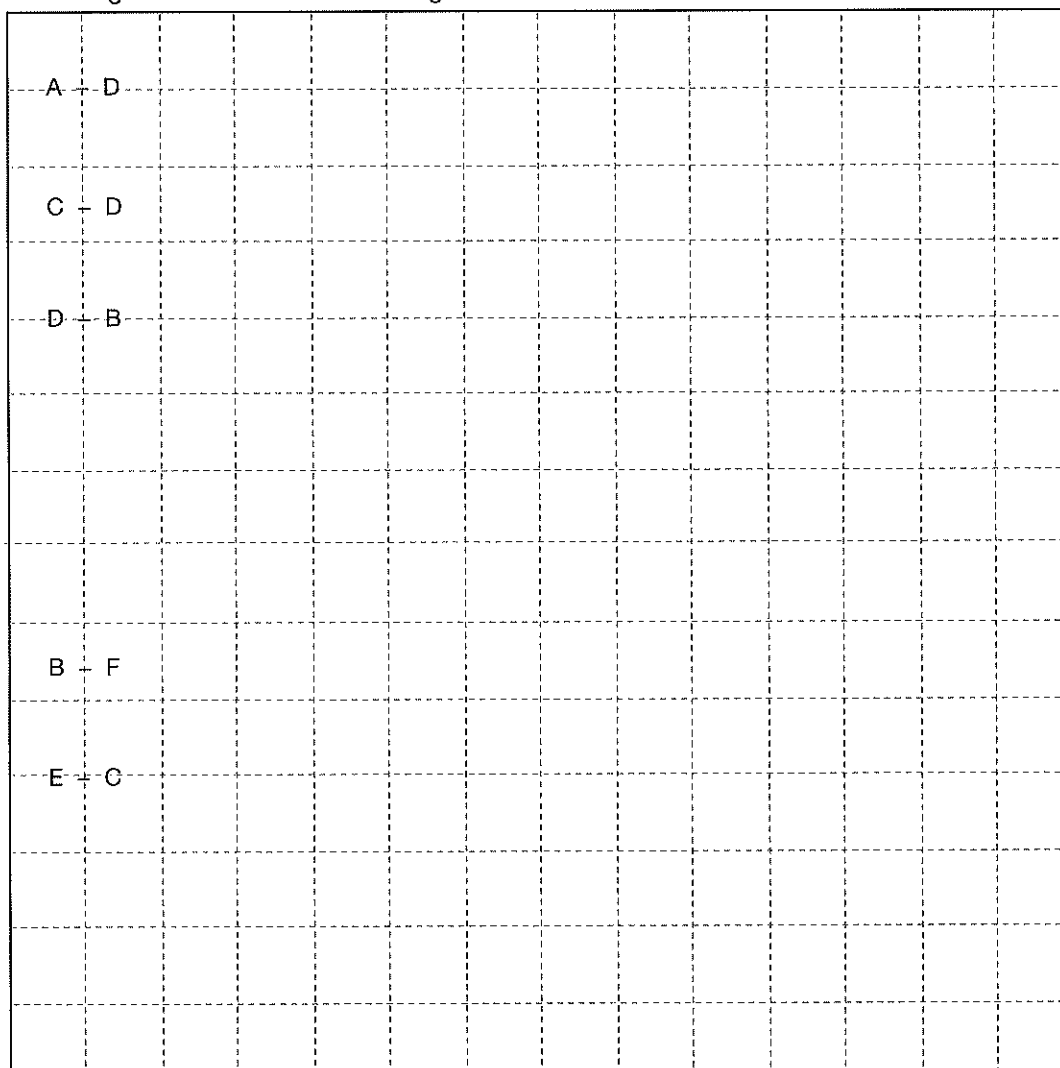
Draw diagrams to show the following vector additions.



1.2.2.6 The diagram represents several displacement vectors.



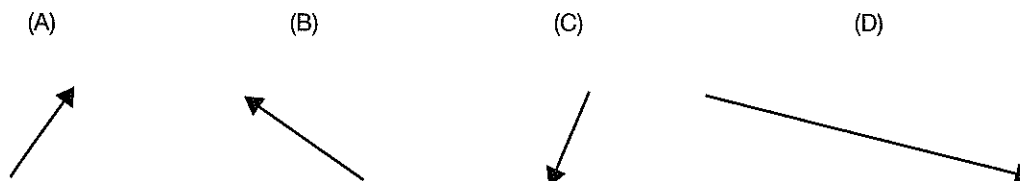
Draw diagrams to show the following vector subtractions.



1.2.2.7 The diagram shows two vectors, P and Q.



Which choice best shows the vector addition $P + Q$?



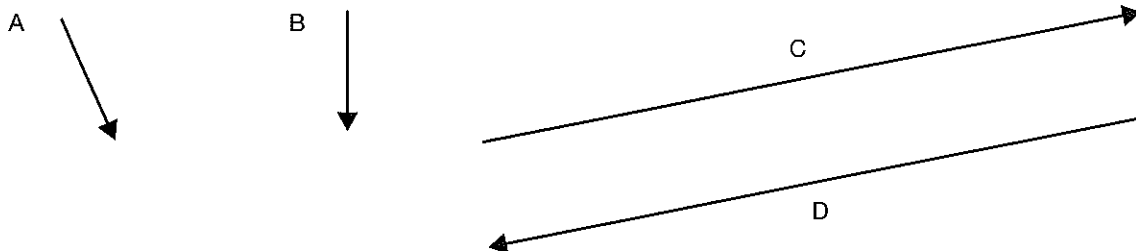
1.2.2.8 A car drives 270 km east from X to Y in 3 hours, and then a further 360 km south to Z in the next 4 hours.

- (a) Calculate its average speed.
- (b) Calculate its average velocity.
- (c) If it drove directly from X to Z at the same average speed, how long would it take?

1.2.2.9 The diagram shows two vectors, X and Y.

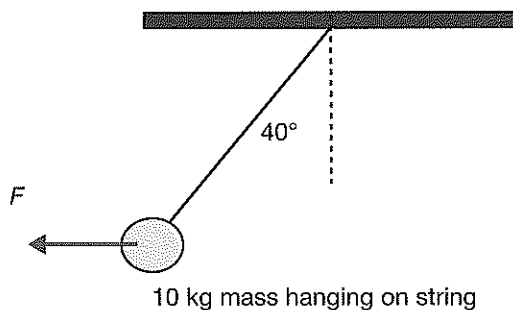


Which choice best shows vector $Z = \text{vector } X + \text{vector } Y$?

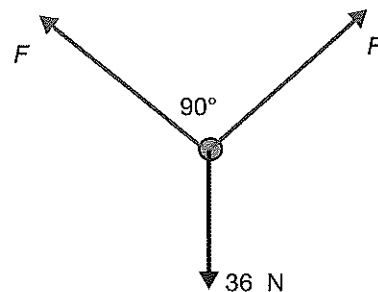


1.2.2.10 Determine the value of the resultant force or the force labelled F in each situation below.

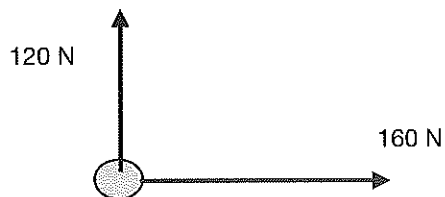
(a) Find the value of F .



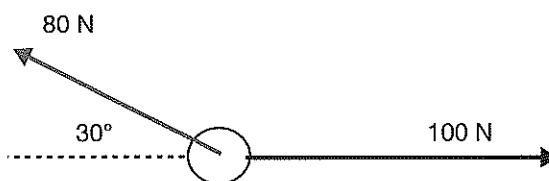
(b) Find the value of F .



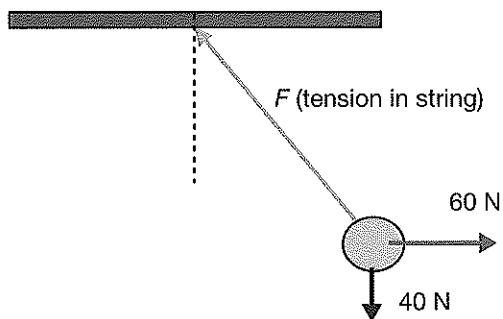
(c) Find the resultant force.



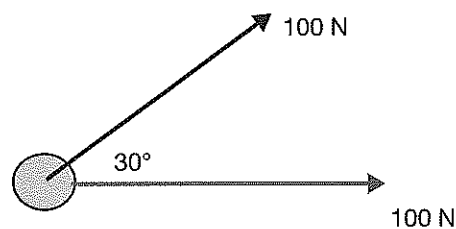
(d) Find the resultant force.



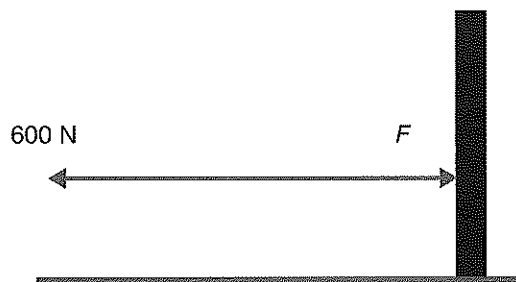
(e) Find the value of F .



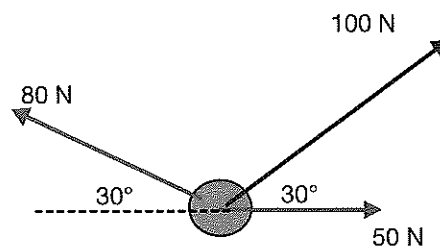
(f) Find the resultant force.



(g) Find the value of F .



(h) Find the resultant force.



1.2.3 More vectors in two dimensions.

1.2.3.1 An object moves from X to Y, 8.0 m E. It then moves from Y to Z, 3.0 m S. From Z it moves 12.0 m W to point P. Calculate its displacement when it is at P.

1.2.3.2 A billiard ball moves north at 5 m s^{-1} , hits a second ball and bounces off at the same speed in an easterly direction. Find its (a) change in speed, (b) change in velocity and, if the change took 0.05 s, (c) its acceleration.

1.2.3.3 A plane flies 1800 km on a bearing of 125° , then turns and flies on a bearing of 215° for a further 2400 km. Calculate its displacement at the end of this flight.

1.2.3.4 A car travels at 30 km h^{-1} N for 45 s then turns to the west and drives at 20 km h^{-1} for a minute.

(a) How far, in metres, does the car travel?

(b) What is its displacement, again in metres, at the end of this time?

1.2.3.5 X walks 6 km E from O, turns, and then walks 8 km bearing 270° . Calculate its displacement.

1.2.3.6 A car travels north at 50 km h^{-1} for 4 hours. It then travels west for 4 hours at 75 km h^{-1} . Calculate its (a) average speed, (b) displacement and (c) its average velocity.

1.2.3.7 Three towns, X, Y and Z are joined by straight roads. XY is 6 km, YZ is 8 km, and XZ is 10 km.

(a) If Y is north of X, what are two possible positions for Z?

(b) What are two possible positions of X relative to Z?

1.2.3.8 The water in a river flows at 0.6 m s^{-1} . A person can swim at 1.5 m s^{-1} . The person dives into the river to swim straight across.

(a) How will the current affect her direction of travel?

(b) How will the current affect her speed through the water?

(c) In what direction should she swim so that she swims straight across the river?

(d) If the river was 100 m wide, how long would this take her?

1.2.3.9 A car travels 120 km east and then north for 60 km. It then travels 40 km on a bearing of 270° . Calculate its final displacement.

1.2.3.10 A vector is 20 cm long. It acts 45° north of east. It has two components at right angles to each other. One is in the direction 33° north of west.

(a) Calculate the direction of the other.

(b) Calculate its magnitude.

(c) Calculate the magnitude of the first component.



1.2.3.11 A person can swim at 1.6 m s^{-1} . She dives in a river and swims across in 25 s, ending up 15 m downstream from where she jumped in.

- (a) How wide is the river?
- (b) At what rate is the river flowing?
- (c) In what direction should she swim in order to reach the other side of the river exactly opposite where she dives in?

1.2.3.12 By drawing a vector diagram, find the resultant displacement of a car, which drives 15 km N, then 10 km W, then 7 km NE, then 6 km N.

1.2.3.13 A car drives 240 km S from P to Q in 6 hours, and then 180 km E to R in the next 4 hours.

- (a) Calculate its average speed.
- (b) Calculate its average velocity.

1.2.3.14 A vector of magnitude 4 units is added to a vector of magnitude 9 units. What are the (a) maximum and (b) minimum values for the resultant?

1.2.3.15 A car travels at 12 m s^{-1} E for 50 s then turns to the left and drives at 16 m s^{-1} for another 50 s.

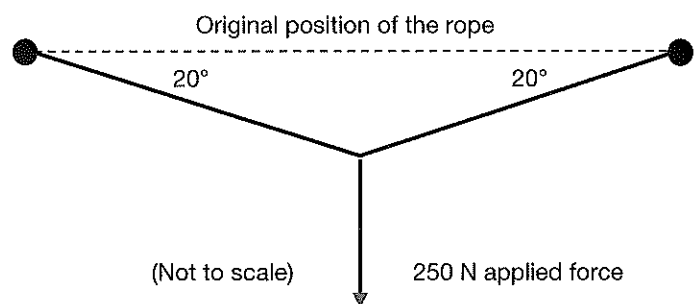
- (a) How far has the car travelled?
- (b) What is its displacement, again in metres, at the end of this time?
- (c) Calculate its average speed.
- (d) Calculate its average velocity.

1.2.3.16 A plane flying at 450 km h^{-1} north encounters a 160 km h^{-1} wind from the east.

- (a) If the pilot makes no adjustments to the controls, in which direction will it fly?
- (b) What will be its new speed?
- (c) In what direction must the pilot head in order to keep flying due north?
- (d) What will be its net velocity in the northerly direction?

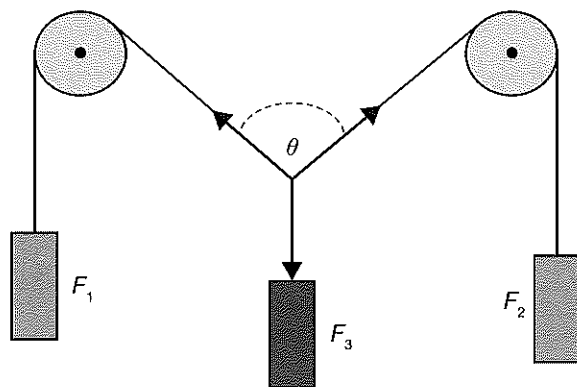
1.2.3.17 A rope is fixed to two posts and pulled from the middle by a force of 600 N. This causes the rope to move so that each end makes an angle of 20° to its original position as shown in the diagram. What is the tension in the rope?

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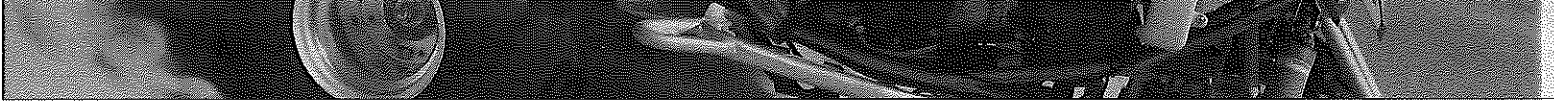
1.2.4 Analysing a vector experiment.

1.2.4.1 A group of students set up the following apparatus to investigate the addition of vectors. They hung masses on mass carriers to produce forces F_1 and F_2 on the ends of a long string, then placed a third mass carrier and masses, F_3 on the string between the pulleys. They adjusted the masses on each carrier until the system was in equilibrium, then measured angle θ . Their results for the three sets of forces are shown in the table.



Run	F_1	F_2	F_3	θ
1	60 g	90 g	132 g	59°
2	40 g	70 g	95 g	68°
3	45 g	55 g	70 g	93°

By drawing appropriate vector diagrams, show that these results are consistent with F_3 being the resultant of F_1 and F_2 .



1.2.5 Graphical representation and analysis of motion.

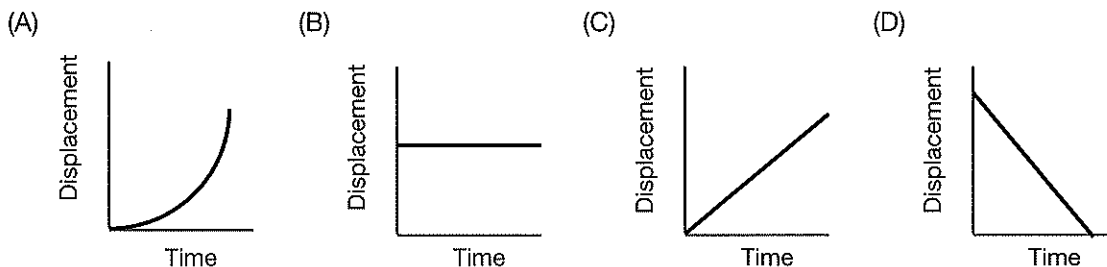
Use the following information to answer the next TWO questions.

In a drag race on a straight car track, the cars accelerate over a distance of 100 m, starting from a zero speed. One car travels the 100 m with a constant acceleration of 0.9 m s^{-2} .

1.2.5.1 What is the time taken by the car to cover the 100 m from the start?

- (A) 9.0 s
- (B) 14.9 s
- (C) 90.0 s
- (D) 222 s

1.2.5.2 Which of the following graphs shows the motion described above?

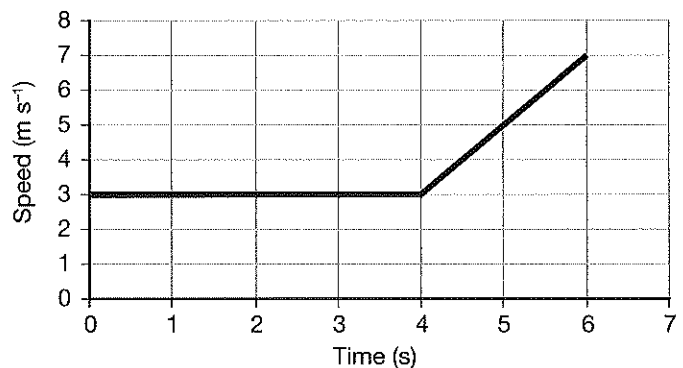


Use the following information to answer the next TWO questions.

The velocity-time graph shows the motion of a car.

1.2.5.3 What total distance is travelled by the car in the 6 s?

- (A) 18 m
- (B) 22 m
- (C) 26 m
- (D) 42 m



1.2.5.4 What is the average speed during the 6 second time interval?

- (A) 3.0 m s^{-1}
- (B) 3.7 m s^{-1}
- (C) 4.3 m s^{-1}
- (D) 7.0 m s^{-1}

Use the following information to answer the next THREE questions.

The diagram shows a graph of the motion of four objects.

1.2.5.5 Which object is moving with constant speed?

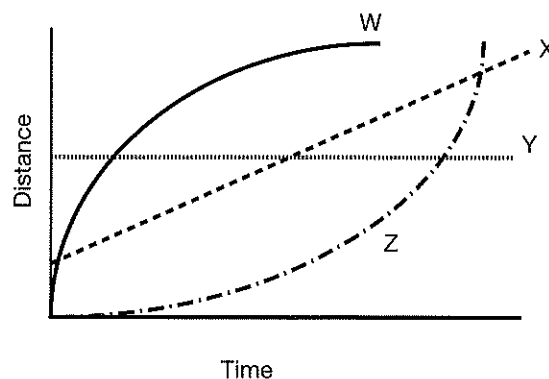
- (A) W
- (B) X
- (C) Y
- (D) Z

1.2.5.6 Which object is stationary?

- (A) W
- (B) X
- (C) Y
- (D) Z

1.2.5.7 Which object is slowing down?

- (A) W
- (B) X
- (C) Y
- (D) Z

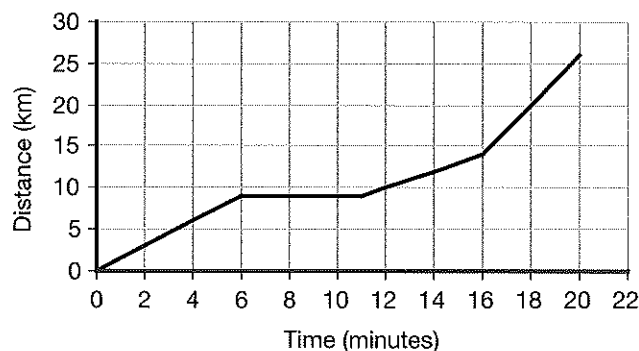


Use the following information to answer the next THREE questions.

The graph represents Tina's journey as she rode her bicycle from a friend's place back to her home.

1.2.5.8 How far is it from Tina's friend's place to home?

- (A) 9 km
- (B) 20 km
- (C) 25 km
- (D) 26 km

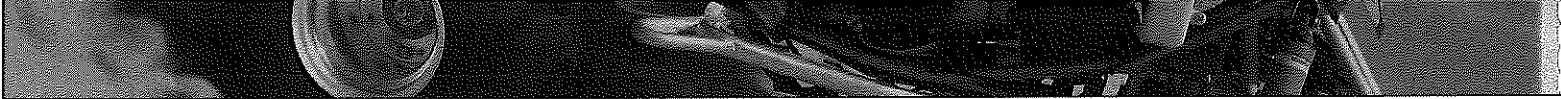


1.2.5.9 Tina stopped for a rest on the way home. For how long did she stop?

- (A) 4 minutes
- (B) 5 minutes
- (C) 9 minutes
- (D) 11 minutes

1.2.5.10 Between which times was Tina moving fastest?

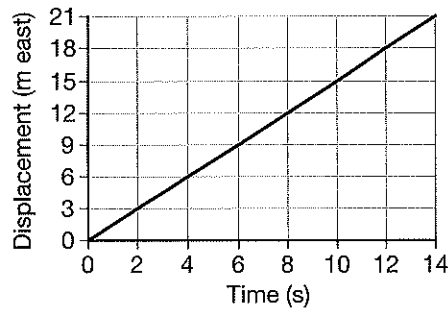
- (A) Between 0 and 6 minutes
- (B) Between 6 and 11 minutes
- (C) Between 11 and 16 minutes
- (D) Between 16 and 20 minutes



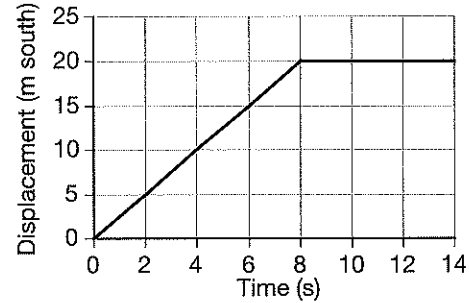
1.2.5.11 The graphs below show how the displacement of several objects varied with time. Describe, in your own words, the motion of each object. Imagine that each of your answers is worth three marks – so you need to say at least three things about the motion of each object. Approach your answers methodically and describe:

- Where the object starts.
- What it does after it starts, including velocity values.
- Where it ends up.

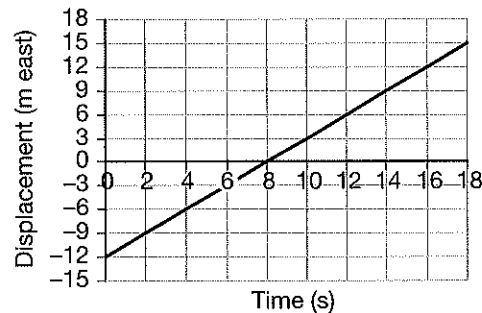
(a)



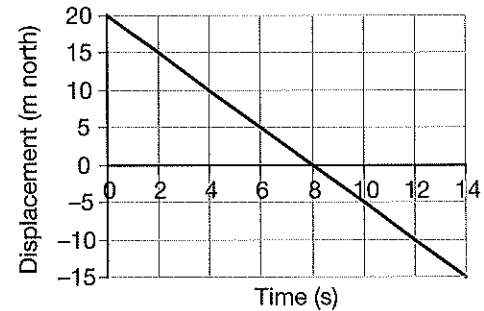
(b)



(c)



(d)



1.2.5.12 Construct, in the spaces below, velocity-time graphs for the graphs from the previous question.

Graph (a)

Graph (b)

Graph (c)

Graph (d)

1.2.5.13 Sketch displacement-time graphs to represent each of the following motions.

(a) A car with constant velocity.

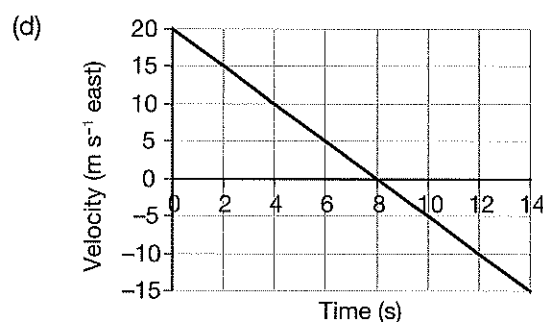
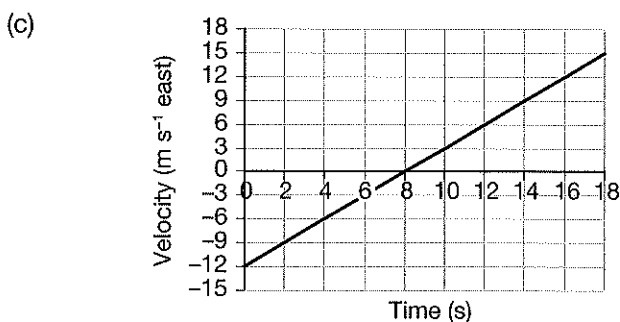
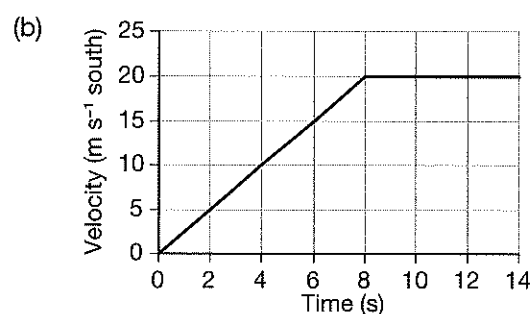
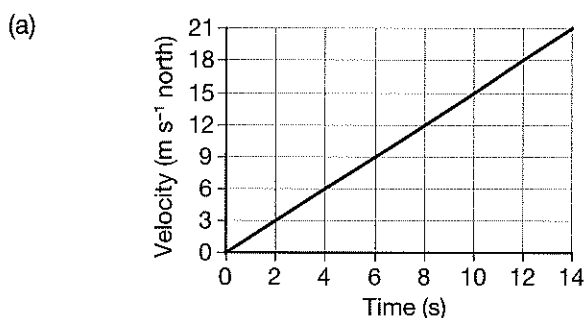
(b) A car with constant acceleration.

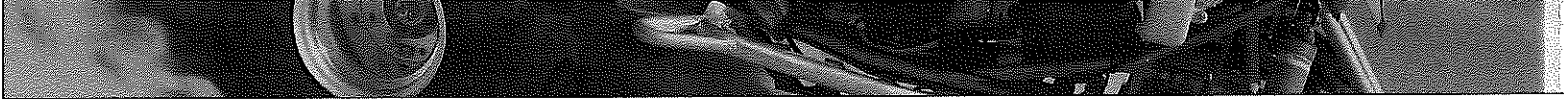
(c) A car which is stationary for a while then decelerates to rest uniformly.

(d) A car which is travelling with constant velocity and then stops.

1.2.5.14 The graphs below show how the velocity of several objects varied with time. Describe, in your own words, the motion of each object. Imagine that each of your answers is worth three marks – so you need to say at least three things about the motion of each object. Approach your answers methodically and describe:

- Where the object starts.
- What it does after it starts, including velocity values.
- Where it ends up.



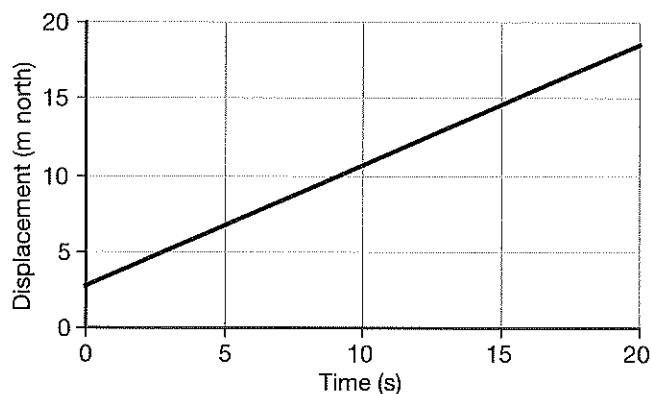


1.2.5.15 Construct, in the spaces below, an acceleration-time graph and a displacement-time graph for the velocity-time graphs in the previous question (assume initial displacement is zero).

Graph	Acceleration-time graph	Displacement-time graph
(a)		
(b)		
(c)		
(d)		

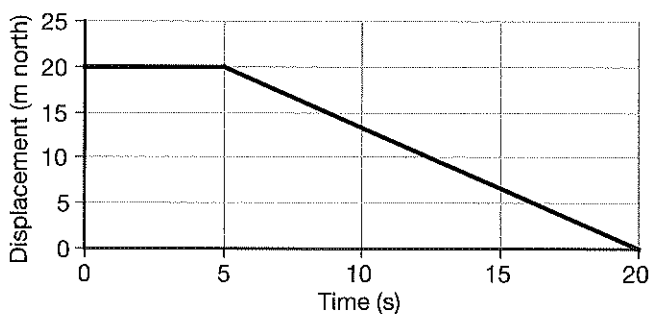
1.2.5.16 Consider the following graph representing the motion of a car.

(a)	How far did the car travel in the first 10 s?	
(b)	Determine its change in displacement.	
(c)	Calculate the average velocity of the car.	
(d)	Find the velocity of the car at time 8 s.	
(e)	Predict the velocity of the car at time 30 s.	



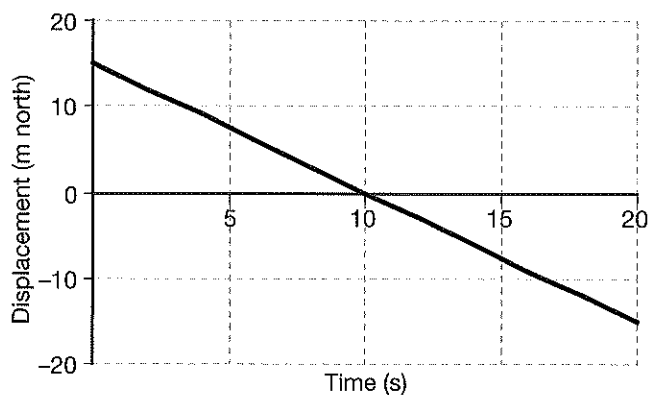
1.2.5.17 Consider the following graph representing the motion of a car.

(a)	How far did the car travel in the first 10 s?	
(b)	Determine its change in displacement.	
(c)	Calculate the average velocity of the car.	
(d)	Determine the velocity of the car at time 8 s.	



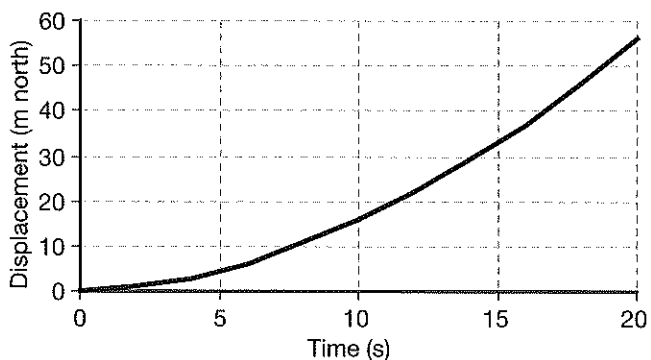
1.2.5.18 Consider the following graph representing the motion of a car.

(a)	How far did the car travel in the first 10 s?	
(b)	Calculate the average speed of the car.	
(c)	Calculate the average velocity of the car.	
(d)	Determine its displacement after 16 s.	
(e)	Determine the velocity of the car at time 8 s.	



1.2.5.19 Consider the following graph representing the motion of a car.

(a)	How far did the car travel in the first 15 s?	
(b)	Determine its total displacement.	
(c)	Calculate the average velocity of the car.	
(d)	Determine the velocity of the car at time 12.5 s.	

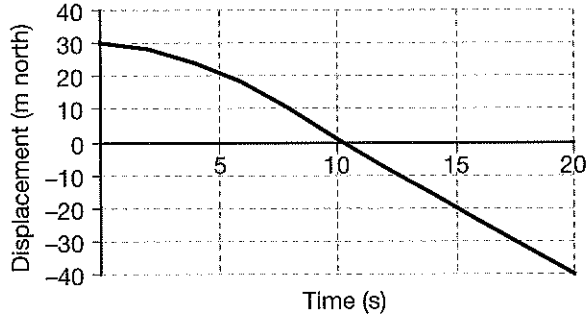


(e) Describe the journey of the car.



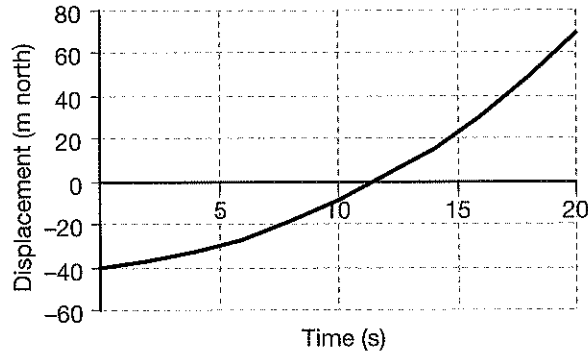
1.2.5.20 Consider the following graph representing the motion of a car.

(a)	How far did the car travel in the first 10 s?	
(b)	Determine its total displacement.	
(c)	Calculate the average speed of the car.	
(d)	Calculate the average velocity of the car.	
(e)	Determine the velocity of the car at time 3 s.	

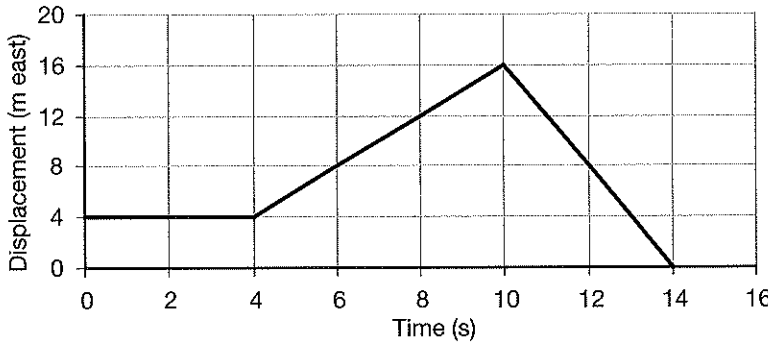


1.2.5.21 Consider the following graph representing the motion of a car.

(a)	How far did the car travel in the first 12 s?	
(b)	Determine its total displacement.	
(c)	Calculate the average speed of the car.	
(d)	Calculate the average velocity of the car.	
(e)	Determine the velocity of the car at time 15 s.	



1.2.5.22 The graph shows how the displacement of a toy car varied over time.



(a) Complete the table to show the displacement of the car at each instant of time given.

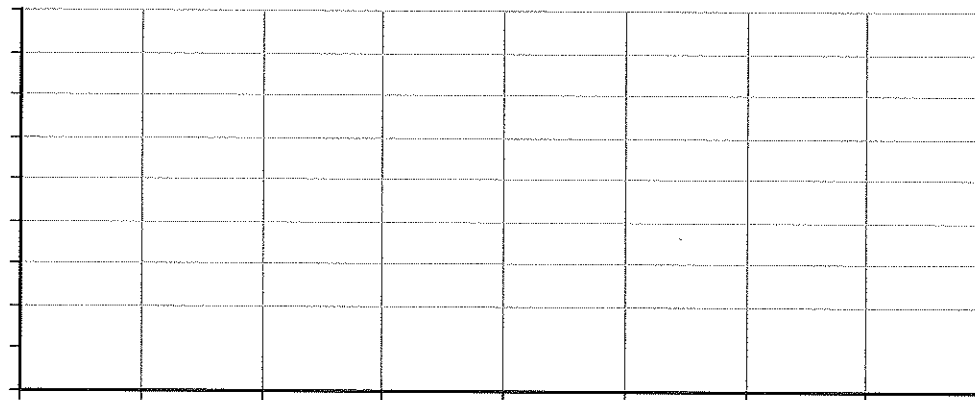
Time (s)	2	4	6	8	10	12	14
Displacement (m east)							

(b) When was its speed constant?
 (c) Calculate the missing values in the table.
 (d) What was the average speed of the car for the whole journey?

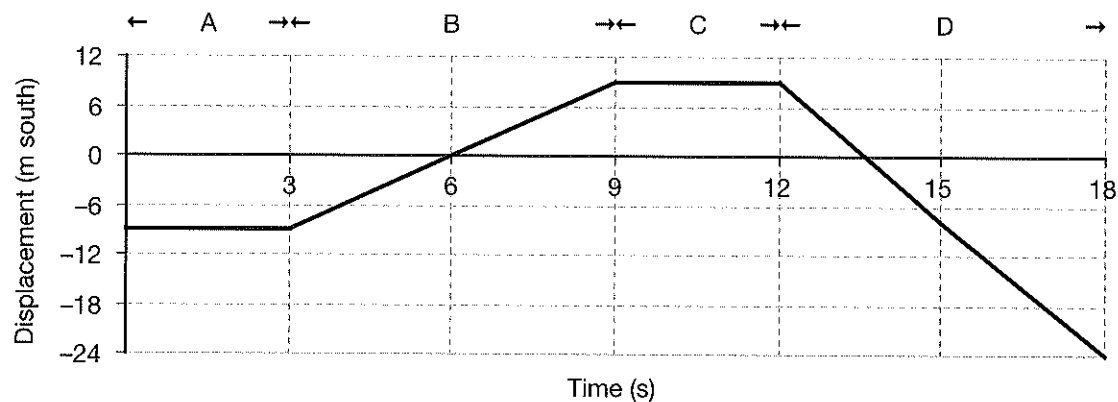
Time interval (s)	Speed during time interval	Velocity during time interval
$t = 0$ and $t = 2$		
$t = 4$ and $t = 10$		
$t = 10$ and $t = 14$		

(e) What was the average velocity for the whole journey?

- (f) Construct, on the grid provided, a distance-time graph for the journey.



1.2.5.23 The graph shows how the displacement of an object changes over time.



- (a) Complete the table to identify the instantaneous velocity of the object at the times indicated.

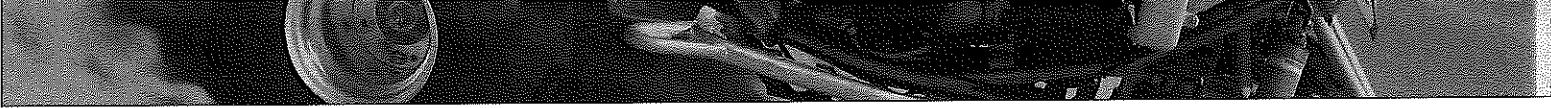
- (b) Complete the table.

Total distance travelled by object	
Total displacement of object	
Average speed of object	
Average velocity of object	

Time (s)	Instantaneous velocity (m s^{-1})
3	
5	
8	
12.5	
16	
18	

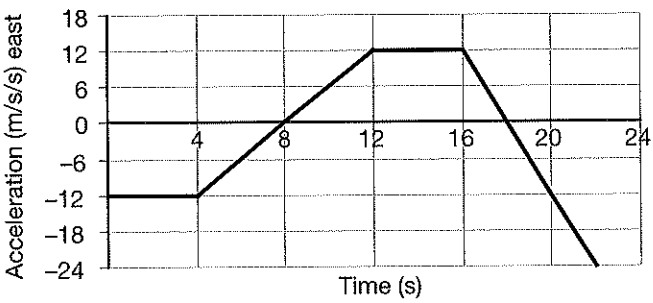
- (c) Complete the following table.

Section	A	B	C	D
How far the object travelled in each section (m)				
Average speed in each section (m s^{-1})				
Average velocity in each section (m s^{-1})				



1.2.5.24 The graph shows how the acceleration of an object changes over time. The object was initially at rest.

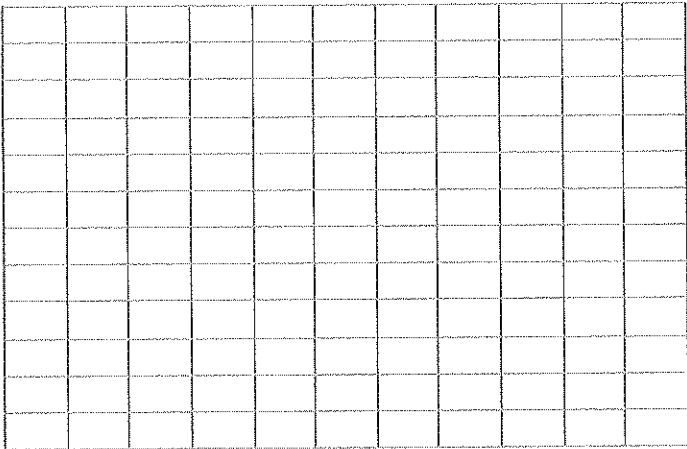
- (a) Complete the table to identify the instantaneous acceleration and the instantaneous velocity of the object at the times indicated.
- (b) Determine the final velocity of the object.



- (c) Determine the change in velocity of the object between $t = 4$ s and $t = 12$ s.
- (d) Determine the change in velocity of the object between $t = 4$ s and $t = 18$ s.

Time (s)	Instantaneous acceleration (m s^{-2})	Instantaneous velocity (m s^{-1})
3		
6		
8		
13		
17		
20		
21		

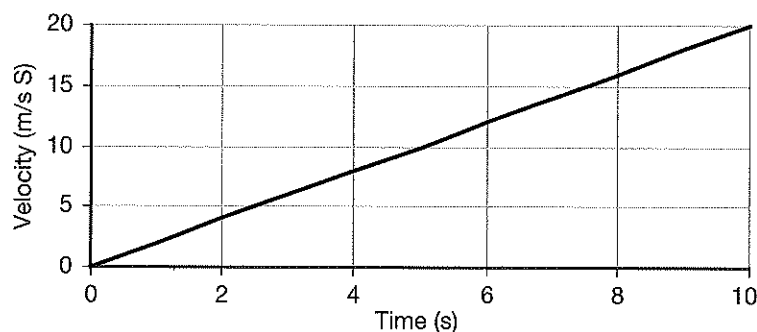
- (e) Construct a velocity-time graph for the object over the 22 s of its motion.



- (f) From your graph, calculate the following quantities.

Total distance travelled by object	
Total displacement of object	
Average speed of object	
Average velocity of object	

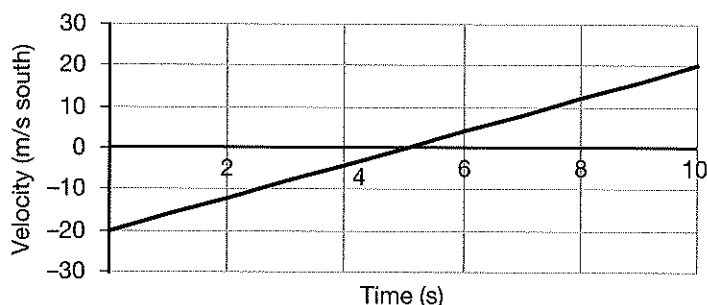
- 1.2.5.25** The graph shows how the velocity of a toy car changes over time. Complete the table by extracting information from the graph.



(a)	Initial velocity of the car.	
(b)	Final velocity of the car.	
(c)	Find the velocity of the car at $t = 7$ s.	
(d)	When was the car moving at 15 m s^{-1} ?	
(e)	What was the acceleration of the car?	
(f)	Change in displacement of the car.	
(g)	Total distance travelled by the car.	
(h)	Average speed of the car.	
(i)	Average velocity of the car.	

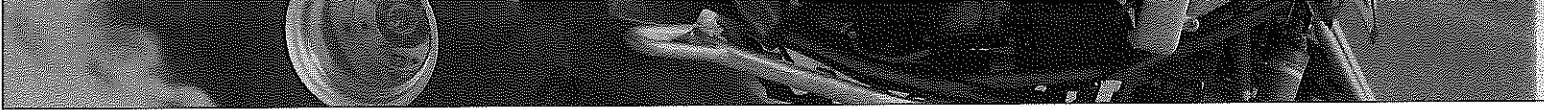
- (j) In (f) you were asked for the change in displacement of the car. Suggest a reason the question did not ask 'what is the displacement of the car?'

- 1.2.5.26** The graph shows how the velocity of a toy car changes over time.

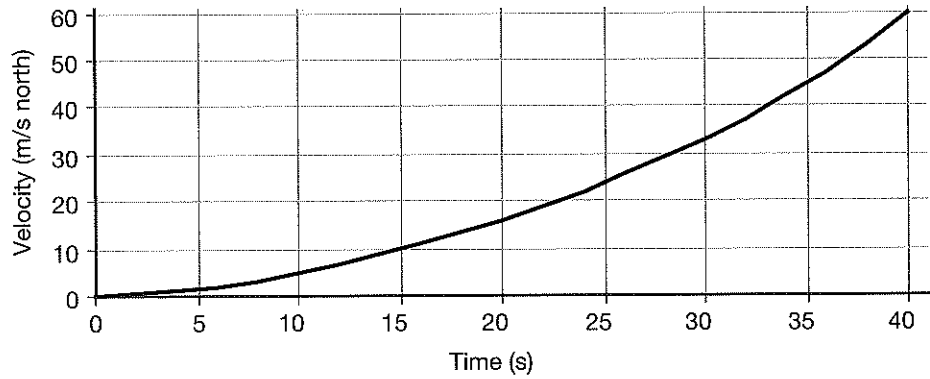


Complete the table by extracting information from the graph.

(a)	What was the initial velocity of the car?	
(b)	What was the final velocity of the car?	
(c)	What was its velocity at $t = 7.0$ s?	
(d)	At what time was its speed 16 m s^{-1} ?	
(e)	What was its acceleration?	
(f)	How far did it travel in the first 5 s?	
(g)	What total distance did it travel?	
(h)	What was its change in displacement?	
(i)	Calculate its average speed.	
(j)	Calculate its average velocity.	



1.2.5.27 The graph shows how the velocity of another car changes over time.



Complete the table by extracting information from the graph.

(a)	What was the initial velocity of the car?	
(b)	What was the final velocity of the car?	
(c)	What was its velocity at $t = 30.0$ s?	
(d)	At what time was its speed 45 m s^{-1} ?	
(e)	What was its acceleration at $t = 20$ s?	
(f)	How far did it travel in the first 25 s?	
(g)	What total distance did it travel?	
(h)	What was its change in displacement?	
(i)	Calculate its average speed.	
(j)	What was its acceleration at $t = 35$ s?	



1.3 Vertical motion.

1.3.1 Measurement of acceleration due to gravity.

1.3.1.1 Analysing an experiment 1.

Distance object falls (m)	Time to fall (s)	(Time to fall) ² (s ²)
1.0	0.58	
2.0	0.82	
3.0	1.00	
4.0	1.15	
5.0	1.29	
6.0	1.41	
7.0	1.53	
8.0	1.64	

The information in the table was obtained from an experiment where the time it took an object to fall from different heights was measured. Use the information to answer the questions.

- (a) Considering the analysis you are asked to do below, suggest a purpose for this experiment.

.....
.....

- (b) Identify three factors which would have been controlled in this experiment.

.....

- (c) Identify the factors that varied.

- (d) Identify the dependent and independent variable in the experiment.

.....

- (e) Identify a possible systematic error in this experiment and suggest a way to minimise its effect.

.....

.....

- (f) Identify a possible random error in this experiment and suggest a way to minimise its effect.

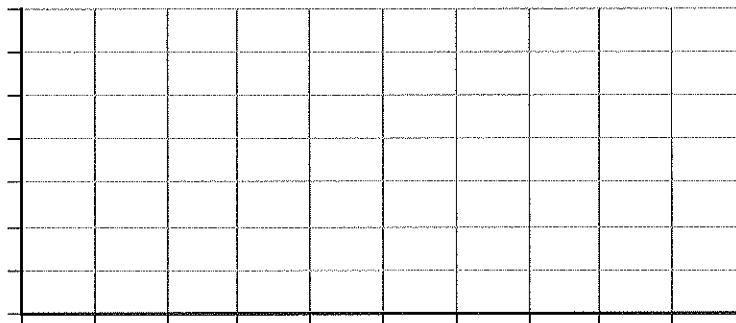
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- (g) Calculate values for t^2 and complete the table above.

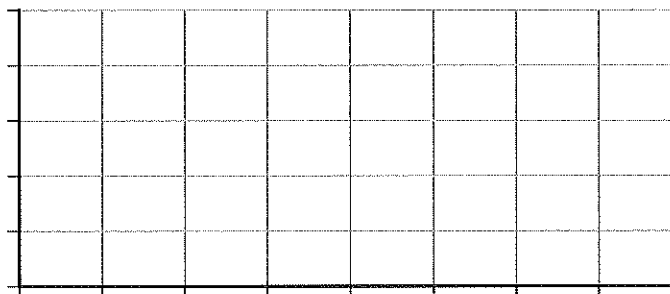


- (h) Construct a graph of the information on the grid. Put t on the x-axis.



- (i) How long would it take to fall 0.5 m?
- (j) How long would it take to fall 3.25 m?
- (k) If it took 2.00 seconds to hit the ground, predict the height the object fell from.
- (l) Discuss whether or not you can write a conclusion from this graph. Explain your answer.

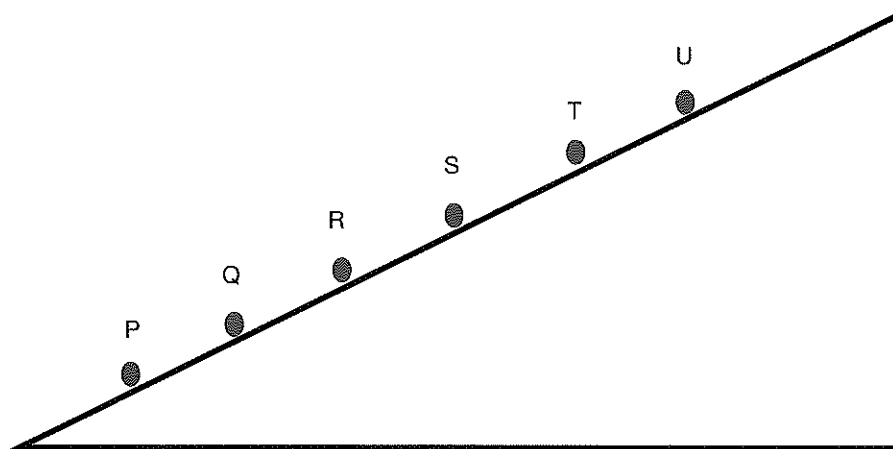
- (m) Construct a graph of distance fallen against t^2 on the grid. Put t^2 on the x-axis.



- (n) What does this graph tell you?
- (o) Given that twice the gradient of your graph is equal to the acceleration of the falling object, calculate this acceleration.
- (p) Deduce whether or not this object is falling on Earth. Explain your answer.
- (q) It is possible to write a conclusion from this graph. Explain why.
- (r) Write a conclusion for the experiment.
- (s) Suggest at least one way this experiment could be improved and explain why this would improve the experiment.

1.3.1.2 Analysing an experiment 2.

In an experiment, a ball was rolled down a ramp from rest, from various positions as shown in the diagram.



The results in the table were obtained. For an object starting from rest, the acceleration is given by the following equation.

$$a = \frac{2r}{t^2}$$

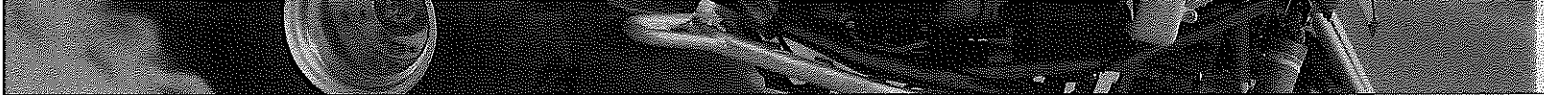
The acceleration due to gravity can be found from an experiment like this by using the following equation.

$$a = g \sin \theta$$

Where θ = angle of incline

(a) Complete the table by making appropriate calculations for the value of the accelerations.

Starting position	Distance up slope (m)	Average time to roll down (s)	(Average time) ² (s ²)	Acceleration of ball down slope (m s ⁻²)
P	0.5	0.55		
Q	1.0	0.82		
R	1.5	1.00		
S	2.0	1.21		
T	2.5	1.30		
U	3.0	1.38		
Average value of acceleration of ball				

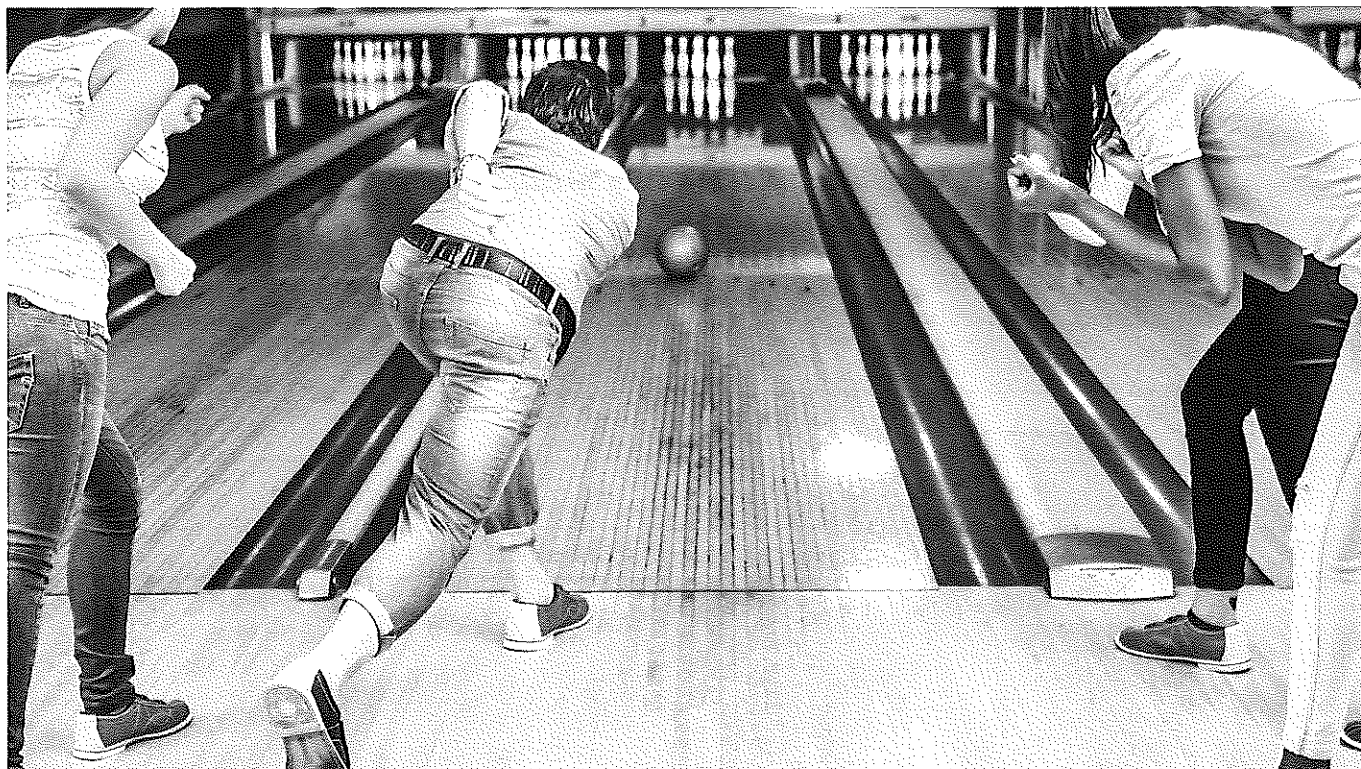


- (b) Graph the distance of the ball up the slope (y-axis) against the $(\text{time})^2$ it takes to roll down the slope.

- (c) Use your graph to find another value for the acceleration of the ball down the slope.
- (d) Which of the two values you have for the acceleration of the ball is the most reliable? Explain your answer.

- (e) Use your results, and the second equation above to find the angle of the incline used in the experiment.

- (f) Assuming that both experiments were done using identical measuring devices, which of the two experiments here (Analysing an experiment 1 and Analysing an experiment 2) would be the more valid? Explain your answer.



1.3.2 Vertical motion near the Earth's surface.

For all questions in this section, take the acceleration due to gravity as 9.8 m s^{-2} , and ignore air resistance unless otherwise stated.

1.3.2.1 A hot air balloon is stationary, 200 m above the ground when an object is dropped vertically from it.

- (a) For how long does the object fall?
- (b) What will be its speed just before it hits the ground?

1.3.2.2 A car, at rest, accelerates at 3 m s^{-2} for 12 s. Calculate its velocity after:

- (a) 4 seconds
- (b) 8 seconds
- (c) 12 seconds

1.3.2.3 A rock falls from rest. Its speed when it hits the ground 3 s later is 29.4 m s^{-1} . Calculate its acceleration.

1.3.2.4 A car is dropped from rest from a height of 50 m in an experiment to examine its safety features. Determine:

- (a) How long it takes to hit the ground.
- (b) Its speed just before it hits the ground.
- (c) Its average speed while falling.

1.3.2.5 A 4.0 kg ball is dropped from the top of 147 m high vertical cliff.

- (a) How long does it take to reach the ground at the bottom of the cliff?
- (b) What will be its speed just before it hits the ground?
- (c) How will your answers to (a) and (b) be different if the mass of the ball is 2 kg? Explain why.

- (d) How far does the ball fall in the first 2.5 s of its fall?

1.3.2.6 A crash dummy was dropped from rest from a hot air balloon in an experiment to examine what would happen to a person crashing into a solid wall at 60 km h^{-1} .

- (a) Express 60 km h^{-1} in m s^{-1}
- (b) Find the height from which it was dropped.
- (c) Find its average speed while falling.

1.3.2.7 A ball is dropped from rest from a hot air balloon. It hits the ground 20 s later.

- (a) Calculate the speed with which it hits the ground.
- (b) Find the height it was dropped from.
- (c) Calculate its average speed while falling.



- 1.3.2.8** A ball is thrown vertically up at 68.6 m s^{-1} .
- (a) How long will it take to reach its maximum height?
 - (b) What will be its speed at its maximum height?
 - (c) What will be its maximum height?
 - (d) How long will it take to fall from the maximum height back to its original level?
- 1.3.2.9** An object is thrown vertically up at 49 m s^{-1} .
- (a) How long will it be before it returns to its launch position?
 - (b) How high will it rise?
 - (c) What will be its height above the launch position at time 3 s ?
 - (d) When will it be 50 m above the launch point?
- 1.3.2.10** A hot air balloon is rising at 2.35 m s^{-1} . When it is 24.5 m above the ground, the pilot drops a sand bag vertically downwards.
- (a) What will be the speed of the bag just before it hits the ground?
 - (b) How long does it take for the bag to hit the ground?
 - (c) The balloon rises at the same rate after the bag is dropped. What will be its altitude as the bag hits the ground?
-
- 1.3.2.11** A stone is thrown vertically up from the ground at 14.7 m s^{-1} near a tree which is 19.6 m high.
- (a) What is its speed at its highest point?
 - (b) How long does it take to reach the highest point?
 - (c) How much higher than the tree does it rise?
 - (d) What will be its speed when it is half way back to the ground?
- 1.3.2.12** Object A is dropped from rest from the top of a 150 m high cliff.
- (a) How long before A hits the ground?
 - (b) What is the speed of A just before it hits the ground?
- 1.3.2.13** An arrow is fired vertically up at 49.0 m s^{-1} .
- (a) What will be its time of flight?
 - (b) For what period of time will it be at least 100 m above its firing height?
 - (c) What will be its maximum height?
 - (d) What will be its acceleration at its maximum height?
- 1.3.2.14** State the average value of the acceleration due to gravity near the Earth's surface.

1.3.2.15 Identify factors which cause the acceleration due to gravity at the surface of the Earth to vary from the value given in your answer to the previous question. Outline how each factor affects the value.

1.3.2.16 Object A was dropped from rest from the top of a 394 m high cliff.

How long after this should object B be thrown vertically down at 8.0 m s^{-1} so that they both hit the ground at the same time?

1.3.2.17 Object A is thrown vertically upwards from the edge of a 58.8 m high cliff at 19.6 m s^{-1} . It rises, then falls to the ground at the bottom of the cliff. Object B is dropped vertically downwards from rest at the instant that A is thrown upwards.

- (a) How long does A take to reach its maximum height?
- (b) How far does A rise?
- (c) What total distance does A fall?
- (d) How long does it take A to fall from maximum height to the ground?
- (e) What is the total time of flight for A?
- (f) What is the speed of A just before it hits the ground?
- (g) How long does B take to hit the ground?
- (h) How long after B does A hit the ground?

1.3.2.18 An object falls from a height of 30 000 m towards the surface of Earth.

Is it correct to say that its acceleration is 9.8 m s^{-2} downwards? Explain your answer.

1.3.2.19 There is another factor which affects the rate of fall of an object apart from the value of the acceleration due to gravity.

Identify this factor and state its effect.

1.3.2.20 An object is dropped from a stationary hot air balloon and hits the ground moving at 127.4 m s^{-1} .

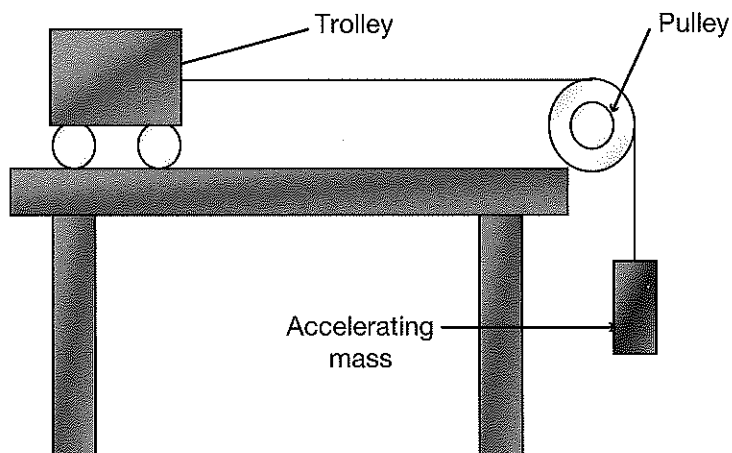
- (a) For long did it fall?
- (b) What was the height of the hot air balloon when the object was dropped?

1.4 Newton's three laws of motion.

1.4.1 Analysing some motion experiments.

1.4.1.1 Analysing an experiment 1.

In this experiment, a trolley of mass 1.5 kg was placed on the bench accelerated by a force attached to it by a string, which passed over a pulley. The mass of the trolley was kept constant and different forces were used to accelerate it from rest across the benchtop. The time it took the trolley to travel 1.0 m across the benchtop was measured. The results are shown in the table.



Run	Accelerating force (N)	Time to travel 1.0 m (s)	Initial speed of trolley (m s^{-1})	Average speed of trolley (m s^{-1})	Final speed of trolley (m s^{-1})	Acceleration of trolley (m s^{-2})
1	F	0.89				
2	$2F$	0.63				
3	$3F$	0.52				
4	$4F$	0.45				
5	$5F$	0.40				

- (a) Complete the table by calculating values for all missing data.
- (b) Draw a graph to show the relationship between the force and acceleration produced. (Place force on the y-axis.)

- (c) Identify the dependent and independent variable in this experiment.

Dependent variable =

Independent variable =

- (d) How could you determine if this experiment was reliable or not?
-
- (e) If it was not reliable, what could you do to try and make it reliable?
-
-
- (f) Write a conclusion for this experiment.
-

1.4.1.2 Analysing an experiment 2.

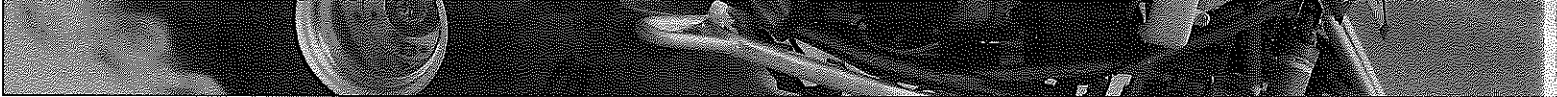
The students did another experiment using the same apparatus. This time they kept the accelerating force constant, and changed the mass of the trolley. They measured how long it took the trolley to move 1.0 m across the benchtop.

Run	Trolley mass (kg)	Time to travel 1.0 m (s)	Initial speed of trolley (m s^{-1})	Average speed of trolley (m s^{-1})	Final speed of trolley (m s^{-1})	Acceleration of trolley (m s^{-2})	(Acceleration) $^{-1}$
1	1.0	1.83					
2	2.0	2.58					
3	3.0	3.16					
4	4.0	3.65					
5	5.0	4.08					

- (a) Complete the table of results by calculating values for all missing quantities.
- (b) Graph mass versus (acceleration) $^{-1}$.
- (c) Use your graph to write a conclusion for the experiment.
-
-
-

Combining your conclusions for experiments 1 and 2:

- (d) Use the conclusions you have written to write a mathematical equation, which shows the relationship between the *mass* of an object, the *force* acting on it and the *acceleration* the force produces.
-
- (e) Use the results of the experiments to find the value of the accelerating forces in the two experiments above.
- Experiment 1
- Experiment 2



1.4.2 Newton's first law of motion.

1.4.2.1 State Newton's first law of motion.

1.4.2.2 Describe an example of a situation which represents Newton's first law of motion.

1.4.2.3 Explain why it is dangerous to drive a car without wearing a seatbelt.

1.4.2.4 Explain why it is dangerous to leave loose objects on the back shelf of a car.

1.4.2.5 With reference to the cartoon below, explain the relationship between Newton's first law of motion and inertia.



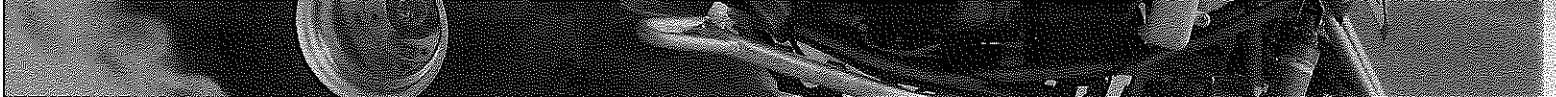
1.4.2.6 Evaluate the relative effectiveness of the seatbelt harness worn by racing car drivers and the lap-sash belts worn by drivers in 'normal' cars.

1.4.2.7 Manufacturers of airbags indicate that young children should not sit in the front passenger's seat of a car if airbags are fitted. Explain why this could be unsafe for small children but safe for adults.

1.4.2.8 Explain the role of the inertia of passengers in each of the following situations. For each situation, outline what forces do act on the passenger (ignore weight force and reaction to weight force).

- (a) The school bus turning a corner.
.....
.....
.....
- (b) A plane accelerating down the runway.
.....
.....
- (c) A lift in a high rise building starting to move upwards.
.....
.....
- (d) A lift in a high rise building starting to move downwards.
.....
.....





1.4.3 Newton's second law of motion.

1.4.3.1

(a) State Newton's second law of motion in terms of forces acting on objects.

.....

.....

.....

(b) State Newton's second law of motion in terms of the momentum of an object.

.....

(c) State the mathematical equation representing Newton's second law.

1.4.3.2 Describe an example of a situation which represents Newton's second law of motion.

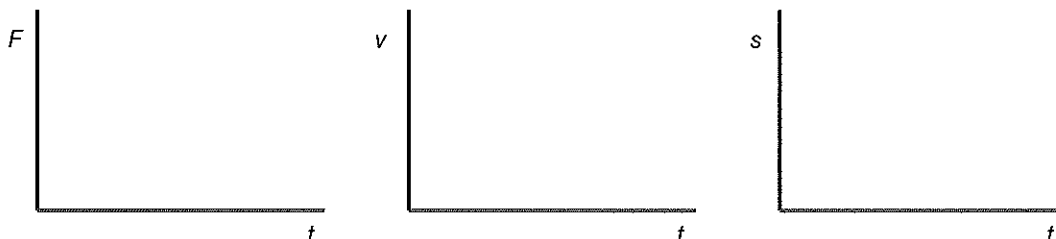
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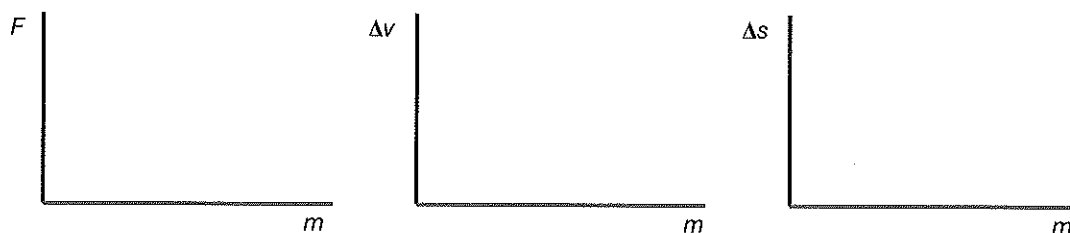
1.4.3.3 Use the formula to find the missing quantities in the table.

Force applied (N)	Mass of object (kg)	Acceleration produced (m s^{-2})
12	3	A =
24	6	B =
150	7.5	C =
45	D =	5
27	E =	3
18	F =	4.5
G =	7	3
H =	1.5	5
I =	0.5	14

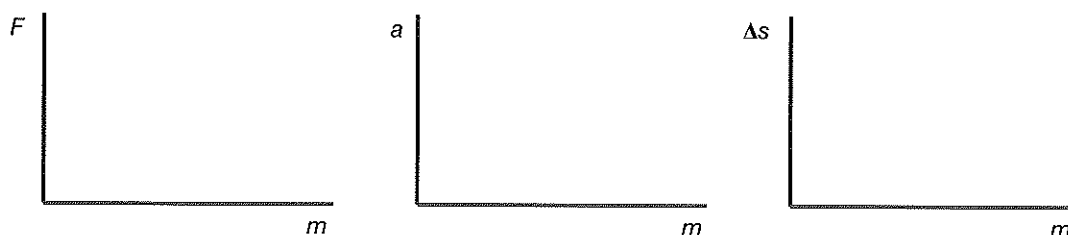
1.4.3.4 A car is accelerating uniformly along a road. Sketch graphs on the axes below to show the relationships indicated.



- 1.4.3.5** Forces act on several cars of different masses to produce identical accelerations. Sketch graphs on the axes below to show the relationships indicated.



- 1.4.3.6** Forces act on several cars of different masses to produce identical velocity changes. Sketch graphs on the axes below to show the relationships indicated.



- 1.4.3.7** Identify what is needed to change the direction or speed of an object.

- 1.4.3.8** What is an object doing if its speed is changing?

- 1.4.3.9** Identify three changes a force acting on a vehicle can cause to its motion.

1.4.3.10

- (a) Clarify the terms 'positive' and 'negative' force as they are used in physics.

- (b) Identify three different situations where the acceleration of a car would be considered to be positive.

- (c) Identify three different situations where the acceleration of a car would be considered to be negative.

- 1.4.3.11** Calculate the force acting on a 5.0 kg mass which accelerates for 3.0 s. During this time its velocity changes from 2.5 m s^{-1} east to 17.5 m s^{-1} west.

- 1.4.3.12** A 12 N force acts on a body and accelerates it from rest to 4.5 m s^{-1} south. This takes 15 s. Calculate the mass of the body.

- 1.4.3.13** An 80 g object is initially at rest. A 0.16 N force to the north acts on it for 5 s. Calculate:

- (a) The acceleration produced by the force.

- (b) The final velocity of the object.

- (c) Its displacement after 5 s.



- 1.4.3.14** Calculate the force that will give a toy car of 2.5 kg an acceleration of 6.0 m s^{-2}
- 1.4.3.15** Calculate the force that gives a toy car of 4.0 kg an acceleration of 1.25 m s^{-2}
- 1.4.3.16** A trolley of mass 2.5 kg is accelerating at 12 m s^{-2} . Calculate the force on it.
- 1.4.3.17** Calculate the acceleration of a 2.5 kg trolley if a force of 37.5 N acts on it.
- 1.4.3.18** A force of 13 N acts on a 4.2 kg trolley. Calculate the acceleration produced.
- 1.4.3.19** A force of 36 N acts on a 3 kg trolley. Calculate the acceleration produced.
- 1.4.3.20** Calculate the mass that accelerates at 2 m s^{-2} when a force of 500 N acts on it.
- 1.4.3.21** A force of 64 N gives a car an acceleration of 0.5 m s^{-2} . Calculate its mass.
- 1.4.3.22** A 3.5 kg toy car is accelerating at 4.0 m s^{-2} . Calculate the force acting on it.
- 1.4.3.23** A 250 g car has a force of 0.75 N acting on it. Calculate its acceleration.
- 1.4.3.24** A force of 1.5 N acts on a 300 gram body. Calculate its acceleration.
- 1.4.3.25** Calculate the mass that accelerates at 0.2 m s^{-2} if a force of 42 N acts on it.
- 1.4.3.26** A 2.5 kg object moving at 5.0 m s^{-1} east is acted upon by a force which changes its velocity to 12 m s^{-1} south over a period of 4 s. By drawing an appropriate vector diagram, calculate:

(a) The change in velocity of the object.

.....

(b) The acceleration of the object.

.....

(c) The force which acted on the object.

.....

1.4.3.27 The velocity of an object changes from 6.0 m s^{-1} west to 8.0 m s^{-1} north when a force of 12 N acts on it for 1.25 s. By drawing an appropriate vector diagram, calculate:

(a) The change in velocity of the object.

.....

(b) The acceleration of the object.

.....

(c) The direction of the force which acted on the object.

.....

(d) The mass of the object.

.....

1.4.3.28 A 750 kg car moving at 32 m s^{-1} west hits a wall and stops in 0.02 s.

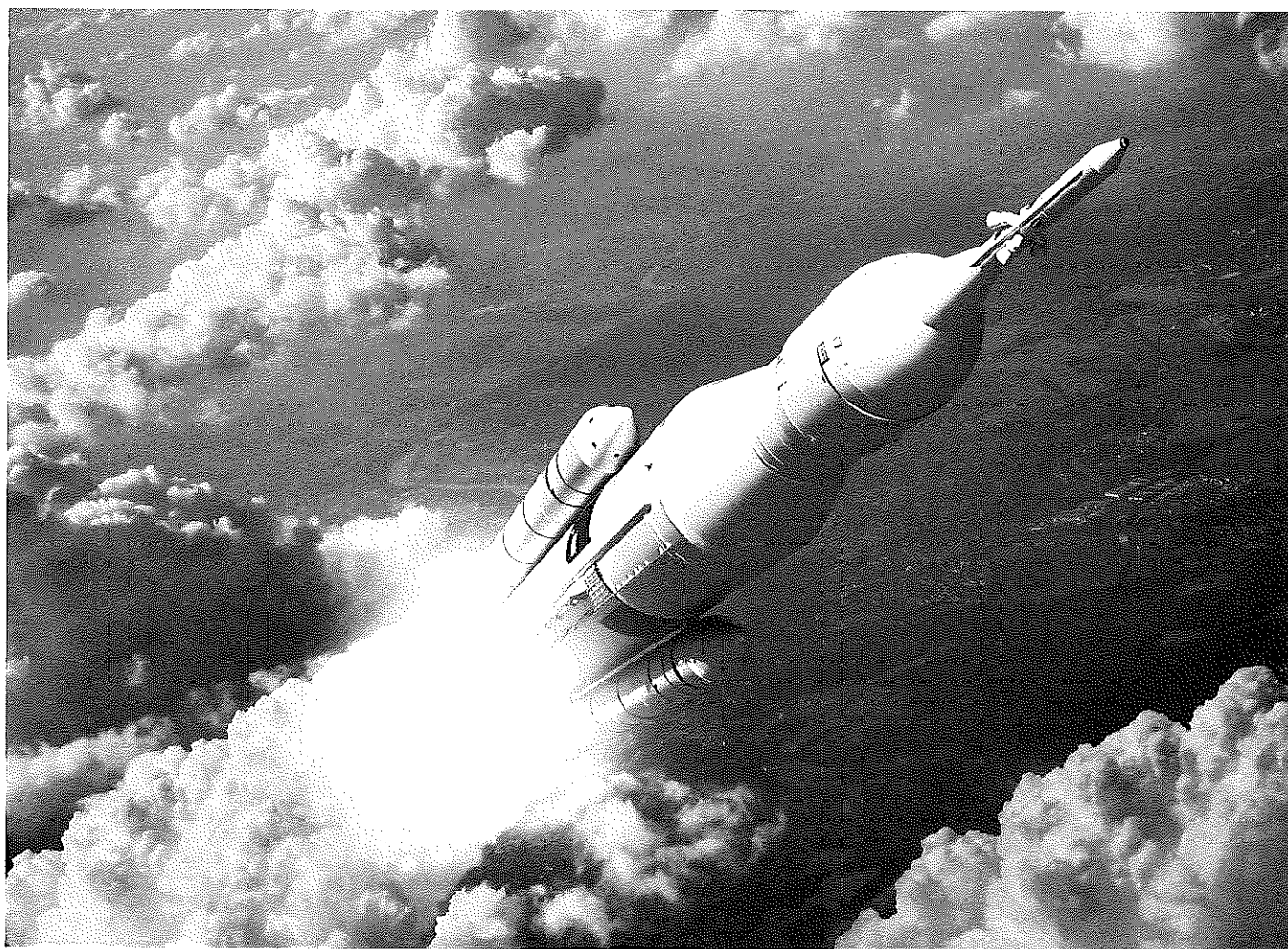
- (a) Calculate the acceleration stopping the car.
- (b) What average force acts on the car to stop it?

1.4.3.29 A 20 000 kg rocket accelerates at 40 m s^{-2} until its speed is 1600 m s^{-1} .

- (a) How long does this take?
- (b) What is the force causing the initial acceleration?
- (c) If the rocket accelerates at 40 m s^{-2} for the whole take-off, then the applied force is not constant. It actually decreases. Explain why.

1.4.3.30 After 4 s of accelerating at 2.5 m s^{-2} , a 500 kg car moves at 50 m s^{-1} east.

- (a) Calculate its initial velocity if the acceleration was positive.
- (b) Calculate its initial velocity if the acceleration was negative.
- (c) Calculate the force acting on the car if the acceleration was positive.
- (d) Calculate the force acting on the car if the acceleration was negative.



1.4.4 Newton's third law of motion.

1.4.4.1 State Newton's third law of motion.

1.4.4.2 Describe an example of a situation which represents Newton's third law of motion.

1.4.4.3 Identify the action and reaction forces in each of the following situations.

(a) A hammer hits a nail on the head.

(b) A tennis racquet hits a ball.

(c) A boy jumps on a trampoline.

(d) A cannon fires a cannonball.

(e) A person walks along the road.

(f) A boxer punches a punching bag.

1.4.4.4 The typical acceleration when a car crashes into a wall at 60 km h^{-1} , or another car is about 150 m s^{-2} . Imagine a mother nursing a 20 kg child in the front seat with no seatbelt on the child.

(a) Calculate the force needed on the child to resist the deceleration of the car.

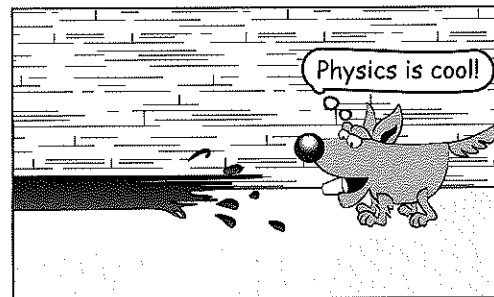
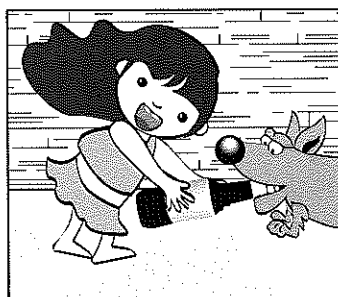
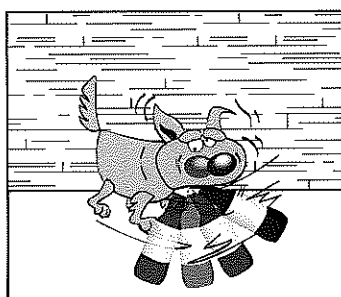
(b) If the mother applied this force to weightlifter's weights, calculate how much she would lift.

(c) Comment on the possibility of the mother doing this.

(d) What implications does this have for nursing an unbelted child in a car?

(e) Identify, and state the law involved in this example.

1.4.4.5 Explain the physics behind the cartoon below.



1.4.5 Using Newton's equations of motion.

1.4.5.1

- (a) Calculate how long it takes a rocket to accelerate from rest to 1200 m s^{-1} if the acceleration is 40 m s^{-2} .
- (b) If the force needed was $25\,000 \text{ N}$, find the mass of the rocket.
- (c) How far does it travel in this time?

1.4.5.2 Calculate how fast a 500 kg rocket would be going if it started from rest and was accelerated by a 5000 N force for 8.75 s .

1.4.5.3 24 N acts on a 0.3 kg ball until it reaches 45 m s^{-1} . If it started at 10 m s^{-1} , calculate how long the force acts.

1.4.5.4 35 N acts on a 5.0 kg mass at rest for 5 s . Find (a) how far it travels and (b) its speed after 5 s .

1.4.5.5 Calculate the force needed to change the velocity of a 9 kg object from:

(a) $12 \text{ m s}^{-1} \text{ W}$ to $27 \text{ m s}^{-1} \text{ W}$ in 4 s .

(b) $8 \text{ m s}^{-1} \text{ N}$ to $16 \text{ m s}^{-1} \text{ S}$ in 8 s .

(c) $10 \text{ m s}^{-1} \text{ S}$ to $35 \text{ m s}^{-1} \text{ N}$ in 2.5 s .

(d) $72 \text{ m s}^{-1} \text{ S}$ to $12 \text{ m s}^{-1} \text{ N}$ in 4 s .

1.4.5.6 6.4 N acts on a body at rest, causing it to reach a speed of 32 m s^{-1} in 9.0 s . Calculate (a) the acceleration of the body and (b) its mass.

1.4.5.7 18 N accelerates a 0.6 kg ball to 60 m s^{-1} .

- (a) If it started at 10 m s^{-1} , calculate how long the force acts.
- (b) How far would it travel while the force was acting on it?

1.4.5.8 Calculate how fast a 2.5 kg toy car would be going if it started from rest and accelerated under the influence of a 15 N force for 0.6 s .

1.4.5.9 A 1.5 kg object is at rest. A 6 N force acts on it for 9 seconds. Calculate (a) how far it travels during the 9 s and (b) its final speed.

1.4.5.10 3.6 N acts on a 0.3 kg toy car until it reaches 5.0 m s^{-1} . If it was initially moving at 0.2 m s^{-1} , find:

- (a) How long the force acts.
- (b) How far the car moves during this time.

1.4.5.11 A bus accelerates uniformly from rest for 9.0 s . It then moves with constant speed for another 45 s over a distance of 900 m . As it approaches the next bus stop it brakes and stops 5 s later.

(a) Find the magnitude of its constant speed.

(b) What is the rate of its acceleration?

(c) What is the rate of its deceleration?

(d) How far is it between the two bus stops?

1.4.5.12 A car accelerates from rest to 4.0 m s^{-1} in 1.2 s .

- (a) What is the acceleration?
- (b) How far does it travel while accelerating?

1.4.5.13 75 N acts on a body at rest of mass 15 kg for 4 s.

- (a) What acceleration is produced?
.....
- (b) What is the speed of the body after 4 s?
.....
- (c) How far does it travel while the force acts?
.....

1.4.5.14 4.5 N acts on a body at rest, causing it to reach a speed of 18 m s^{-1} in 9.0 s.

- (a) Determine its acceleration.
.....
- (b) Calculate the mass of the body.
.....
- (c) How far does it travel in the 9 s?
.....

1.4.5.15

- (a) Find how long it takes a rocket to accelerate from rest to 1200 m s^{-1} if the acceleration is 30 m s^{-2} .
.....
- (b) How far would it travel in this time?
.....
- (c) If the force needed was 25 000 N, calculate the mass of the rocket.
.....

1.4.5.16 A car, moving at 30 m s^{-1} in an 80 km h^{-1} zone passes a stationary police car. The driver of the car brakes urgently to try to hide the fact that he was speeding. He stops in 12 s.

- (a) By how much is the car over the speed limit?
.....
- (b) What is the deceleration of the car?
.....
- (c) How far does the car travel in this time?
.....

1.4.5.17 A car moving at 16 m s^{-1} loses a hub cap. The hub cap rolls for 120 m while the car continues to travel forwards at the same speed.

- (a) What is the acceleration of the hub cap?
.....
- (b) How long does it take to stop?
.....
- (c) How far in front of the hub cap will the car be when the hub cap stops?
.....

1.4.5.18

- (a) Calculate how fast a 25 kg model rocket would be going if it started from rest and a 45 N force was applied to it for 15 s.
.....
- (b) How far would it travel in the first 5 s?
.....
- (c) What would be its velocity after 9 s?
.....

1.4.5.19 A car accelerates uniformly from rest, reaching a speed of 36 m s^{-1} in 2 minutes. Sometime later, it brakes suddenly and stops in 16 s.

- (a) Calculate its acceleration.
.....
- (b) How far does it travel while accelerating?
.....
- (c) What is its acceleration when braking?
.....
- (d) How far does it travel while braking?
.....
- (e) If the speed limit in the area is 100 km h^{-1} , by how much is the car under or breaking the speed limit when it is at maximum speed?
.....

1.4.5.20 A car travelling at 9 m s^{-1} E accelerates at 3.5 m s^{-2} E for 6 s, then at 2.0 m s^{-2} E for a further 4 s.

- (a) Calculate the final velocity of the car.
-
- (b) Determine how far it travels in the 10 s.
-
- (c) At what constant acceleration would it need to travel to reach the same speed in the same time?
-

1.4.5.21 A sports car travelling at 20 m s^{-1} passes a police car which is cruising at 15 m s^{-1} . The police car accelerates at 1.5 m s^{-2} and gives chase.

- (a) How long does it take the police to catch up to the sports car?
-
- (b) Find the distance the police car travels before it catches up to the sports car.
-

1.4.5.22 A bike rider starts from rest and accelerates at 1.4 m s^{-2} for 5.0 s and then continues with uniform speed. He stops with a uniform deceleration over a period of 7.5 s. He travels a total distance of 150 m.

- (a) Find his maximum speed for the journey.
-
- (b) What is the magnitude of his deceleration?
-
- (c) How far does he travel at constant speed?
-

1.4.5.23 A car accelerates from rest to 7.5 m s^{-1} N. During this time it travels 125 m

- (a) What is the acceleration?
-
- (b) For how long does the car accelerate?
-
- (c) It stopped in 60 m. What is the deceleration?
-
- (d) How long would it take to stop?
-

1.4.5.24 A ball rolls down a smooth slope from rest with an acceleration of 4.9 m s^{-2} .

- (a) What will be its speed after 3 s?
-
- (b) How long will it take to reach 24.1 m s^{-1} ?
-
- (c) How far does it travel before it reaches a speed of 24.1 m s^{-1} ?
-
- (d) The slope is inclined at 30° . Derive an expression for the acceleration down a slope in terms of the angle of incline and g , the acceleration due to gravity.
-

1.4.5.25 18 N acts on a body at rest. It reaches 72 m s^{-1} in 4.0 s. Calculate:

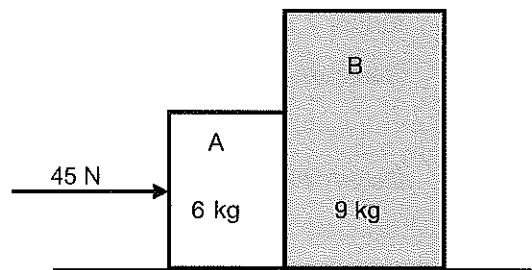
- (a) The acceleration of the body.
-
- (b) Its mass.
-
- (c) How far it moves in the 4 s.
-

1.4.6 Objects in contact.

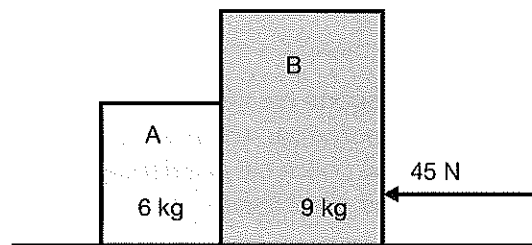
1.4.6.1 The diagram shows two blocks, A and B in contact on a smooth surface. A 45 N force acts on block A, and the surface can be considered frictionless.

Calculate:

- The acceleration of each block.
- The net force on each block.
- The force A applies to B.
- The force B applies to A.

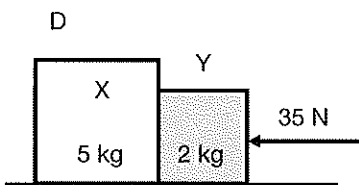
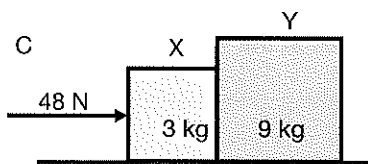
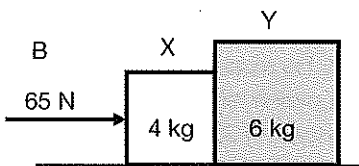
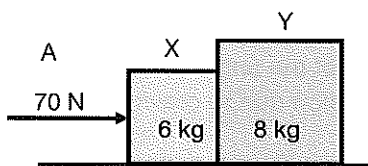


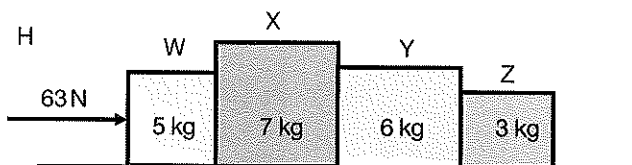
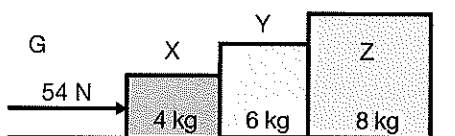
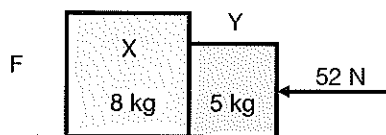
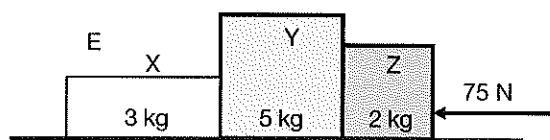
1.4.6.2 Repeat your calculations to answer the same questions for the two block systems shown in the second diagram where the 45 N force is now acting on block B.



1.4.6.3 The diagrams A to H show several systems in contact. Neglect frictional forces. For each system calculate:

- The acceleration of the system.
- The acceleration of each object.
- The net force on each object.
- The value of the action-reaction force pair at each contact surface.

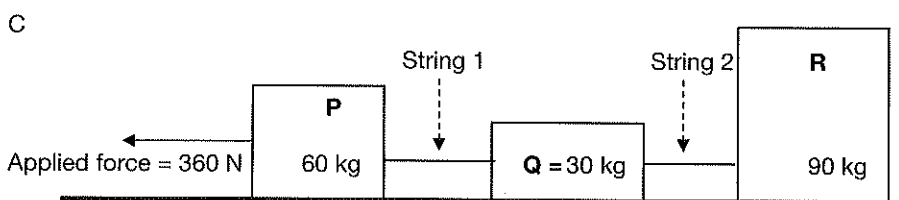
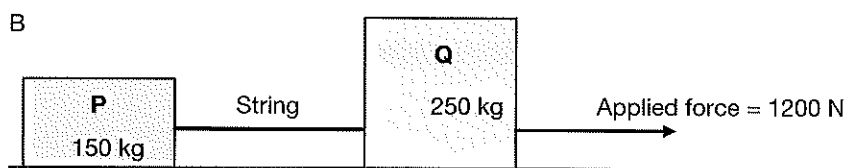
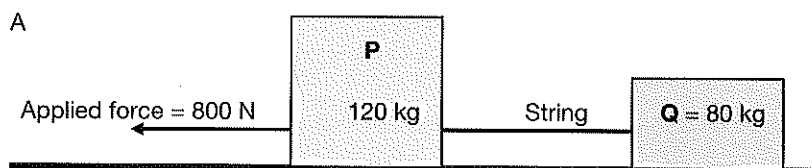


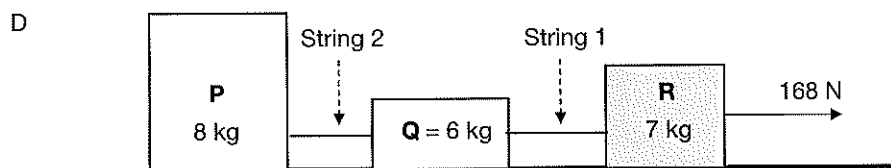


1.4.7 Objects connected by strings.

1.4.7.1 For each of the systems connected by strings shown, neglecting frictional forces, find:

- The acceleration of the system.
- The acceleration of each object.
- The net force on each object.
- The tension in each string or rope.



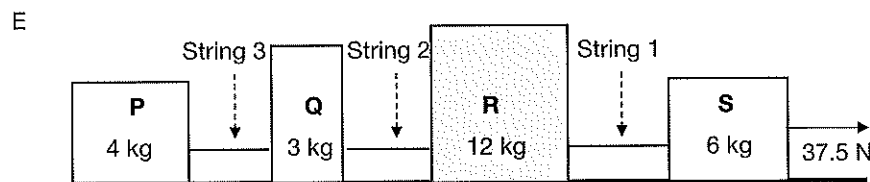


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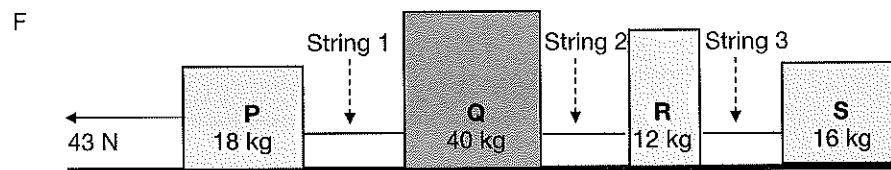


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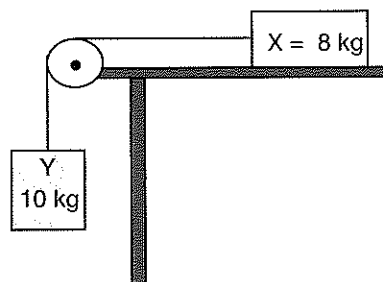
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.....

1.4.7.2 For each of the systems connected by strings shown, neglecting frictional forces, find:

- The acceleration of the system.
- The net force on each object.
- The tension in each string or rope.

A

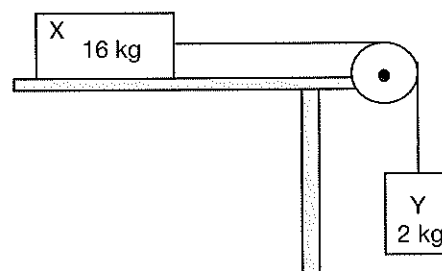


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B

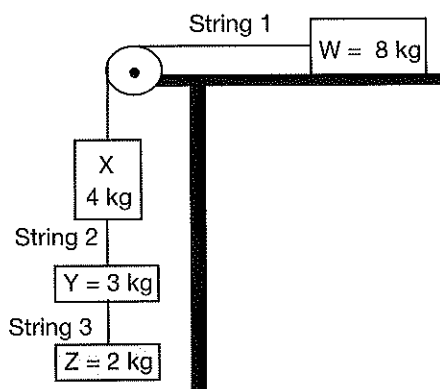


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C

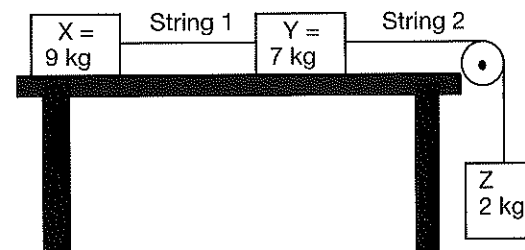


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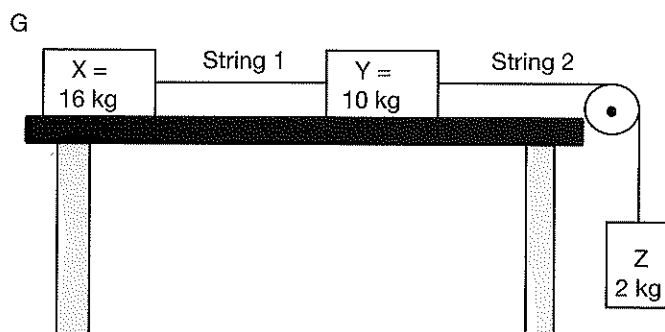
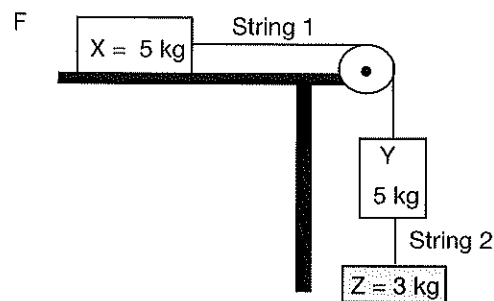
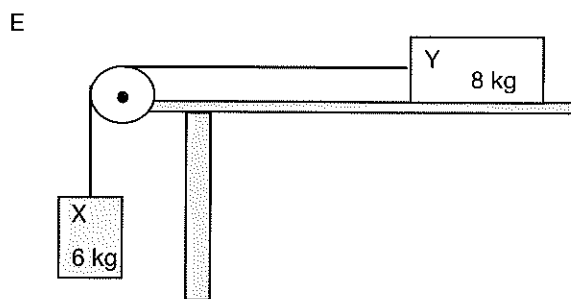
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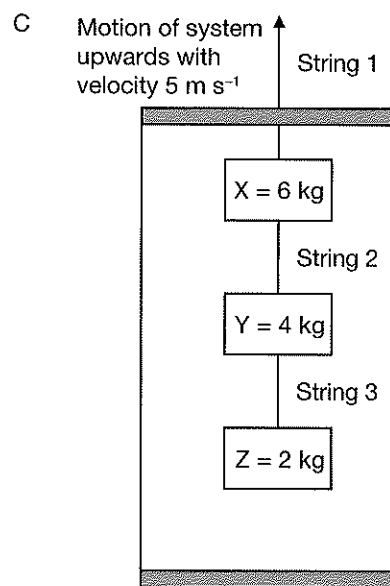
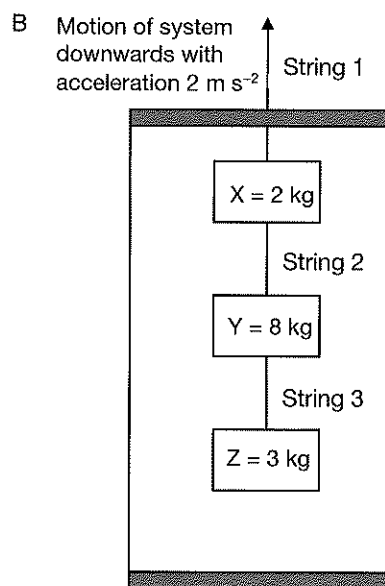
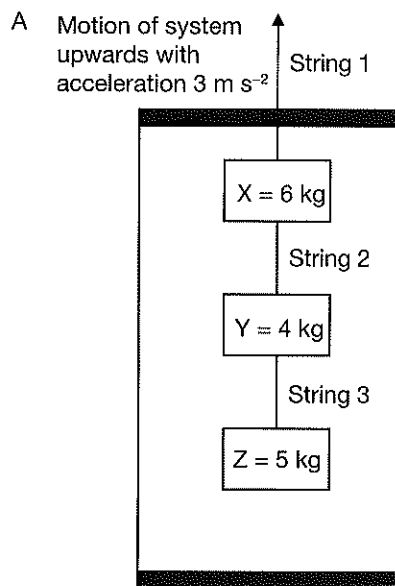
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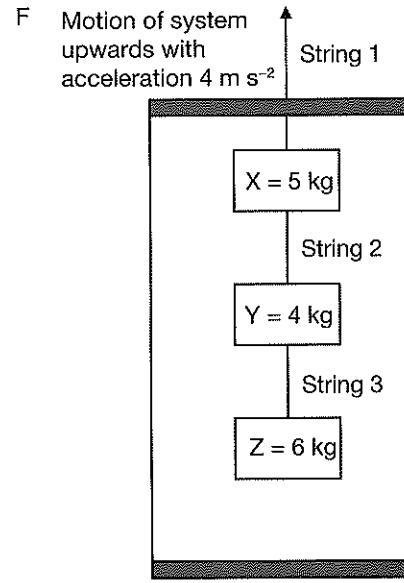
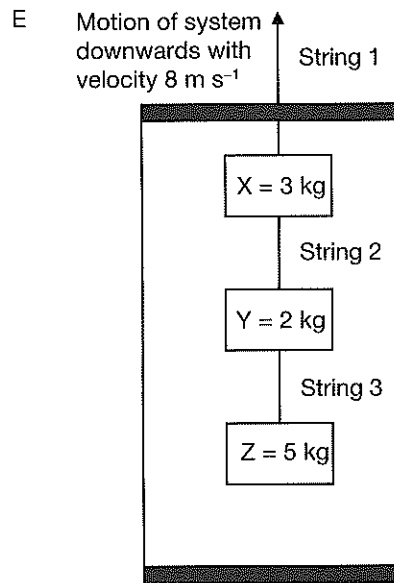
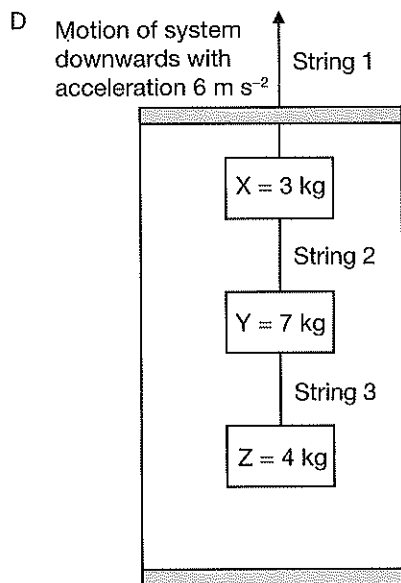
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1.4.7.3 For each of the systems (A to F) connected by strings shown, neglecting frictional forces, find the tension in each string.



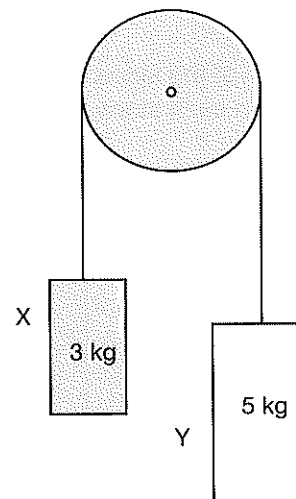


1.4.7.4 Consider the blocks shown in the diagram. They are connected by a light string over a frictionless pulley.

- (a) Label all the force acting in this pulley system.
(b) Calculate the acceleration of the system.

- (c) Calculate the net force on each block.

- (d) Calculate the tension in the string connecting the blocks.



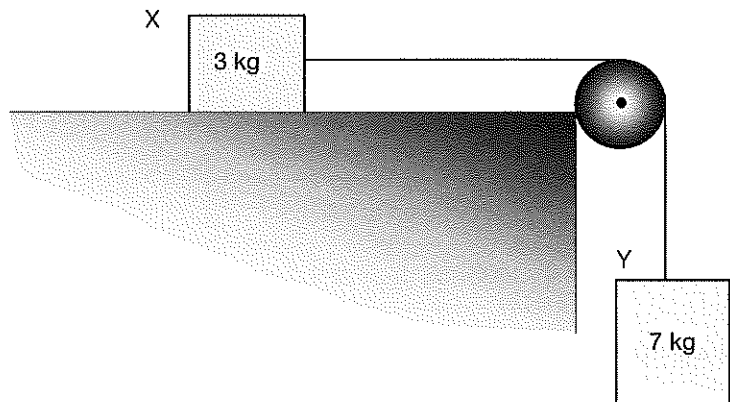
1.4.7.5 A 3 kg mass is resting on a Newton balance in an elevator. Complete the table to show the reading on the balance under the conditions of movement of the elevator given.

	Movement of elevator	Reading on Newton balance
(a)	Stationary	
(b)	Moving up at 2.0 m s^{-1}	
(c)	Moving down at 2.0 m s^{-1}	
(d)	Moving up at 2.0 m s^{-2}	
(e)	Moving down at 2.0 m s^{-2}	

1.4.7.6 A ball is hanging by a string from the ceiling of a bus. Describe the motion of the bus when the string is:

- (a) Hanging straight down.
- (b) Hanging down towards the rear of the bus.
- (c) Hanging down towards the front of the bus.

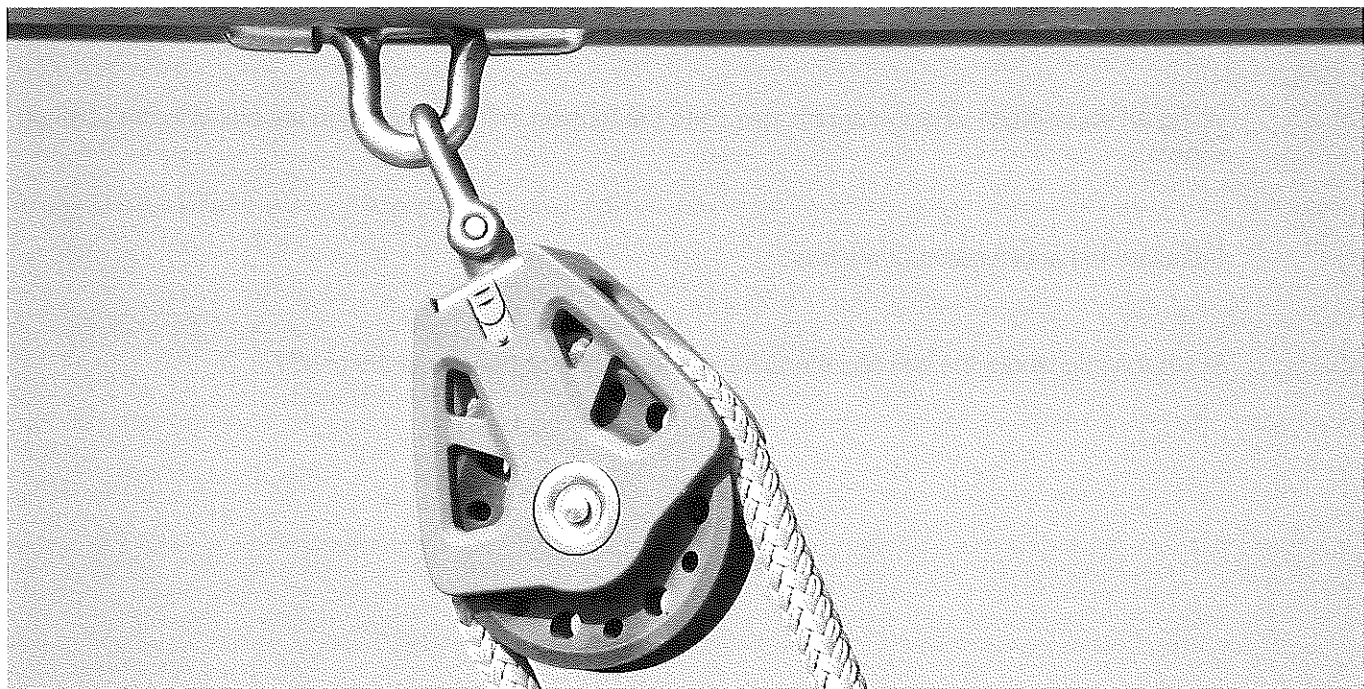
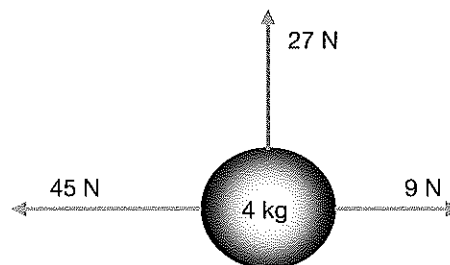
1.4.7.7 Consider the two blocks shown in the diagram. They are connected by a light string over a frictionless pulley. Block X is resting on a smooth surface.



- (a) Label all the forces influencing the motion of this pulley system.
- (b) Calculate the acceleration of the system.
- (c) Calculate the net force on each block.
- (d) Calculate the tension in the string connecting the blocks.

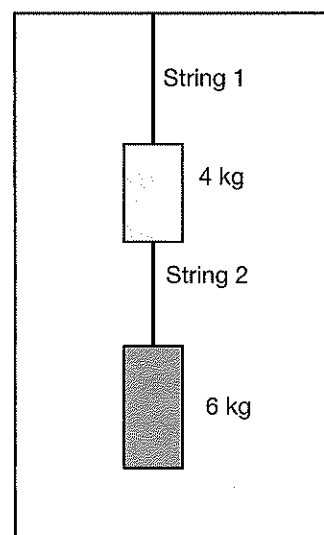
1.4.7.8 Consider the object shown in the diagram. Several forces act on this object which is at rest on a smooth, horizontal surface.

- (a) Draw a vector diagram to find the net force on the object.
- (b) Calculate the acceleration of the object.



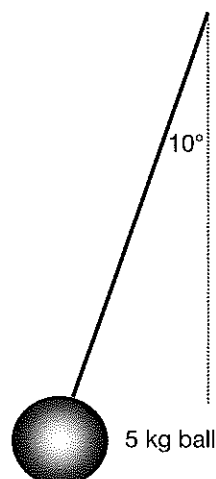
- 1.4.7.9** Two masses are connected by a string and are hanging from the ceiling of an elevator as shown. Complete the table to show the tensions in each string under the conditions of movement of the elevator given.

Movement of elevator		Tension in string 1 (N)	Tension in string 2 (N)
(a)	Stationary		
(b)	Moving up at 3.0 m s^{-1}		
(c)	Moving down at 3.0 m s^{-1}		
(d)	Moving up at 3.0 m s^{-2}		
(e)	Moving down at 3.0 m s^{-2}		



- 1.4.7.10** A 3.0 kg ball is hanging by a string from the ceiling of a bus as shown in the diagram.

- (a) Label the forces acting on the ball.



- (b) Using an appropriate vector diagram, calculate the tension in the string when the ball is in the position shown.

.....

- (c) Calculate the acceleration of the bus.

.....

- (d) At what angle would the ball and string hang if the bus was braking at 2.5 m s^{-2} ?

.....

1.5 Momentum of moving objects.

1.5.1 Momentum and impulse.

1.5.1.1 Define momentum and state its units of measurement.

1.5.1.2 Recall the law of conservation of momentum.

1.5.1.3 Define the impulse of a force.

1.5.1.4 Explain the relationship between momentum and impulse.

1.5.1.5 Show, by manipulating the units, that 1 kg m s^{-1} is equal to 1 N s

1.5.1.6 Describe the conditions under which the momentum of an isolated system will be conserved during a collision.

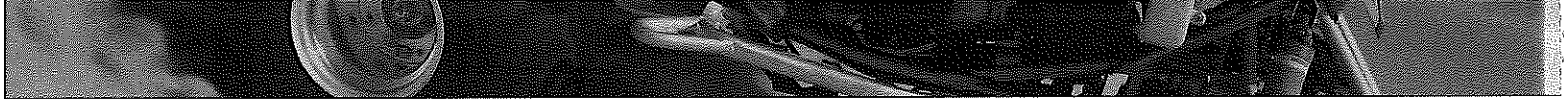
1.5.1.7 Using appropriate equations show how Newton's third law can be used to explain the conservation of momentum during a collision.

1.5.1.8 Calculate the momentum of each of the following.

- (a) A 1600 kg truck moving at 7.0 m s^{-1} east.
- (b) A $5.3 \times 10^{-26} \text{ kg}$ oxygen molecule moving westwards at $1.3 \times 10^3 \text{ m s}^{-1}$
- (c) A 1100 kg car being driven at 72 km h^{-1} south.
- (d) A 2.5 kg bird flying north at 12.0 m s^{-1}
- (e) A 75 kg sprinter running south at 9.5 m s^{-1}

1.5.1.9 Calculate the impulse needed to cause each of the following momentum changes.

- (a) From 10.0 kg m s^{-1} west to 15 kg m s^{-1} west.
- (b) From 0.6 kg m s^{-1} west to 15 kg m s^{-1} east.
- (c) From 12.0 kg m s^{-1} south to 15 kg m s^{-1} north.
- (d) From 10.0 kg m s^{-1} north to 15 kg m s^{-1} north.



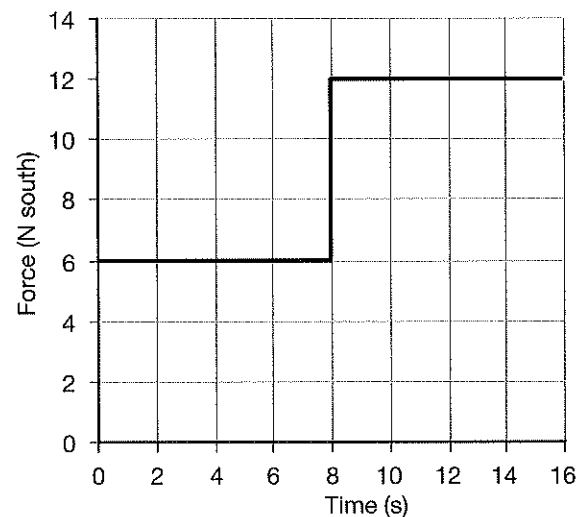
- (e) From 10.0 kg m s^{-1} west to 2.5 kg m s^{-1} west.
- (f) If each of the changes above took 4.0 s , what force was involved? Write your answers next to your answers for (a) to (e).

1.5.1.10 A force of 18 N east acts on a 6.0 kg object for 4.5 s . The object was initially moving westward at 15 m s^{-1} .

- (a) Calculate the impulse of the force.
- (b) Calculate the change in momentum of the object.
- (c) Calculate the final momentum of the object.
- (d) Calculate the final velocity of the object.

1.5.1.11 The graph shows the relationship between the force acting on a 3.0 kg toy car and the time the force acts. The car was initially at rest. Calculate:

- (a) The total impulse applied to the car.

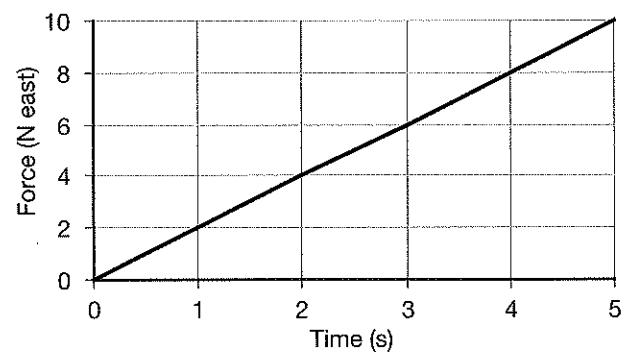


- (b) The final velocity of the object.

1.5.1.12 The graph shows how the force acting on a 0.4 kg object changes with time.

- (a) Calculate the acceleration of the object at $t = 4 \text{ s}$.

- (b) Calculate the impulse applied to the object during the 4 s .



- (c) Calculate the object's change in speed during the 5 s .

- (d) If its initial velocity was 22 m s^{-1} west, calculate its velocity after 4 s

1.5.1.13 A 50 kg passenger in a car is held firmly by her seatbelt during a collision. The seatbelt, in stopping her, puts an average force of $12\,000 \text{ N}$ on her for $1.0 \times 10^{-3} \text{ s}$. Calculate:

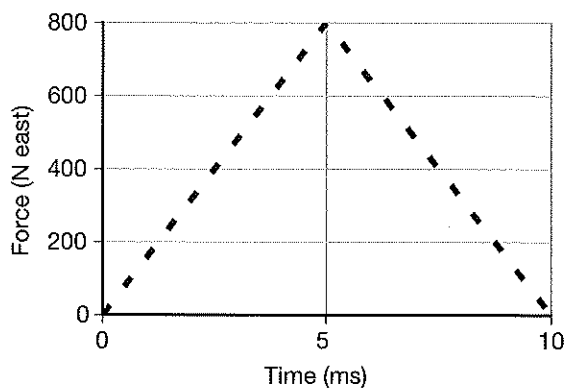
- (a) The impulse of the force.
- (b) The impulse of the force if the collision time was 0.01 s
- (c) The initial speed of the car.

1.5.1.14 A 12 kg ball moving at 2.5 m s^{-1} collides with a brick wall and rebounds at 1.75 m s^{-1} .

- (a) Calculate the change in momentum of the ball.
- (b) Calculate the impulse the ball applies to the wall.
- (c) If the collision lasts 0.15 s, calculate the force the wall places on the ball.

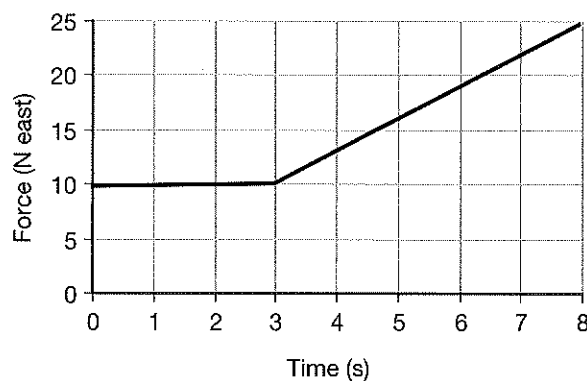
1.5.1.15 The graph shows the force-time relationship for a 50 g golf ball being hit by a club.

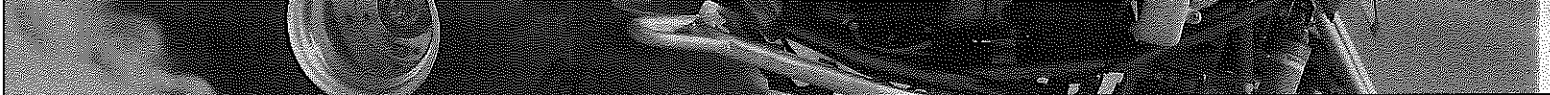
- (a) Identify the maximum force applied to the golf ball.
.....
- (b) Calculate the average force applied to the ball.
.....
- (c) Calculate the total impulse applied.
.....
- (d) Calculate the speed of the ball just after being hit.
.....



1.5.1.16 The graph shows the force on a mass of 2.5 kg over time. Calculate the:

- (a) Impulse applied in the first 2.5 s.
.....
- (b) Impulse applied in the first 5.0 s.
.....
- (c) Mass's change in velocity at 2.5 s.
.....
- (d) Change in velocity at 5.0 s.
.....
- (e) Mass's final velocity if its initial velocity was:
 - (i) Zero
 - (ii) 12 m s^{-1} east
 - (iii) 18 m s^{-1} west





1.5.2 The law of conservation of momentum.

1.5.2.1 A toy car of mass 4.0 kg is struck horizontally by another car causing it to accelerate from 4.0 m s^{-1} to 10.0 m s^{-1} in 0.1 s. If the car was on a frictionless surface, calculate:

- (a) Its change in momentum.
- (b) The impulse applied to it.
- (c) The average force applied to it.

1.5.2.2 A 150 g toy car collides with a padded lounge chair at 0.20 m s^{-1} . The collision takes 0.3 s. The toy car stops after the collision.

- (a) Calculate the impulse the car applies to the lounge.
- (b) Calculate the impulse the lounge applies to the car.
- (c) Calculate the average force the car applies to the lounge.
- (d) Calculate the average force the lounge applies to the car.
- (e) What law are you using to answer questions (b) and (d)?

1.5.2.3 A toy truck, moving eastwards, has a mass of 500 g and hits a stationary toy car at 20.0 m s^{-1} . After impact the truck rebounds at 10 m s^{-1} . Calculate:

- (a) The change in velocity of the truck.
- (b) The change in momentum of the truck.
- (c) The average force on the car if the collision lasted 0.05 s.
- (d) The average force on the car if the collision lasted 0.01 s.

1.5.2.4 A train truck of mass 1500 kg is moving along the track with a constant speed of 0.5 m s^{-1} when it passes under a wheat chute. As it moves under the chute, 2000 kg of wheat drops vertically into it. Calculate the speed of the truck after the wheat has been added.

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1.5.2.5 A 2 tonne truck moving at 15.0 m s^{-1} collides with a concrete retaining wall. The wall does not break. At first appearances this would seem to contravene the law of conservation of momentum. Explain how this collision does not contravene the law of conservation of momentum.

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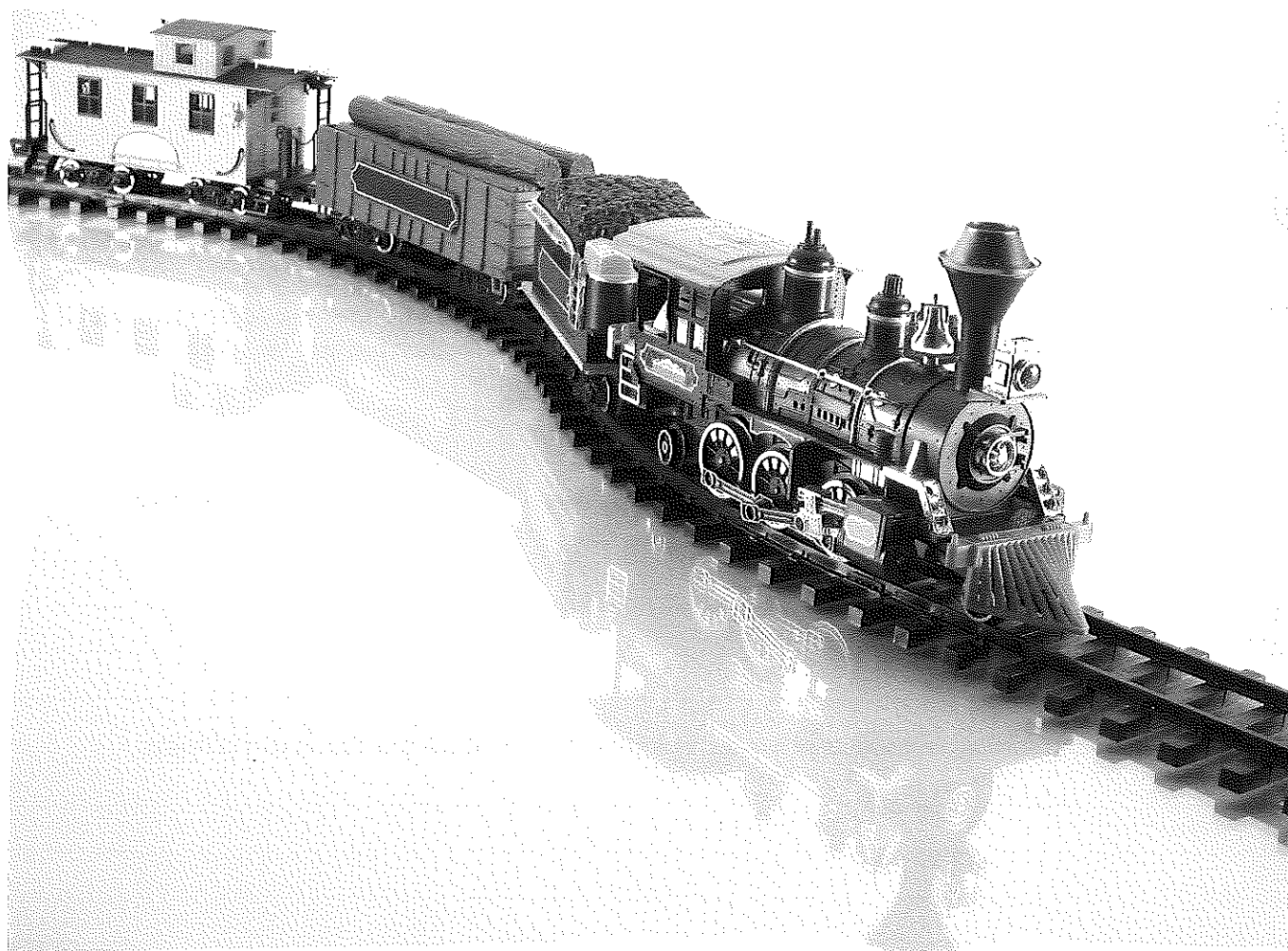
1.5.2.6 A 3.0 kg block, at rest, is hit by a 4.0 kg block moving at 1.5 m s^{-1} . After collision, the 4 kg block stops. Calculate the velocity of the 3 kg block after the collision.

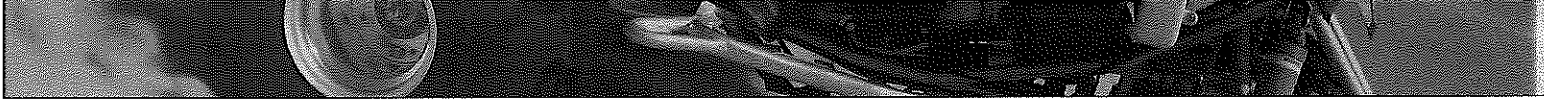
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1.5.2.7 A $6.0 \times 10^3 \text{ kg}$ train engine, moving at 2.5 m s^{-1} collides, and couples with a stationary carriage. They move at 1.2 m s^{-1} . Calculate the carriage's mass.

.....

- 1.5.2.8** A 1200 kg car moving at 60 km h^{-1} collides with an 800 kg parked car. As a result of this collision, the parked car is pushed forwards at 40 km h^{-1} . Determine what happens to the other car.
-
- 1.5.2.9** A 900 kg car, moving at 6.0 m s^{-1} east collides with a 150 kg motorbike moving west at 8.0 m s^{-1} . The car slows to 4.0 m s^{-1} . Calculate what happens to the motorbike.
-
- 1.5.2.10** A 45 kg go-cart crashes into a 30 kg go-cart moving in the opposite direction at 3.0 m s^{-1} . The 30 kg cart rebounds at 4.0 m s^{-1} while the 45 kg cart rebounds at 2.0 m s^{-1} . Calculate the initial speed of the 45 kg go-cart.
-
- 1.5.2.11** A 350 g toy train engine is moving at 12 cm s^{-1} when it collides, and couples with a 200 g carriage moving at 5 cm s^{-1} in the same direction. Calculate their combined speed.
-
- 1.5.2.12** A 400 g toy engine pulls four 150 g carriages. They move at 0.3 m s^{-1} when they run into, and couple with, two identical carriages moving at 0.15 m s^{-1} in the same direction. Calculate the speed of the combination.
-





1.5.3 Momentum and safety during collisions.

1.5.3.1 A passenger in a car that has a synthetic plastic bumper bar will usually suffer less injury than a passenger in a car with a solid, chromed metal bumper bar. Explain how this is possible.

.....

.....

.....

1.5.3.2 With reference to the concepts of impulse and momentum, assess the reasons for the introduction of the following.

(a) Low speed zones in built-up areas.

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(b) The use of airbags in collisions.

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(c) The inclusion of crumple zones in vehicles.

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1.5.3.3 Evaluate the relative safety of the seatbelt harness worn by racing car drivers and the lap-sash belt worn by drivers in 'normal' cars.

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1.5.3.4 Manufacturers of airbags indicate that young children should not sit in the front passenger's seat if airbags are fitted. Predict why this could be unsafe for small children but safe for adults.

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1.5.3.5 Explain the idea of crumple zones in cars and how they work to reduce injury to passengers.

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1.5.3.6 Recall and assess the benefits of five technologies used to reduce the effects of a collision.

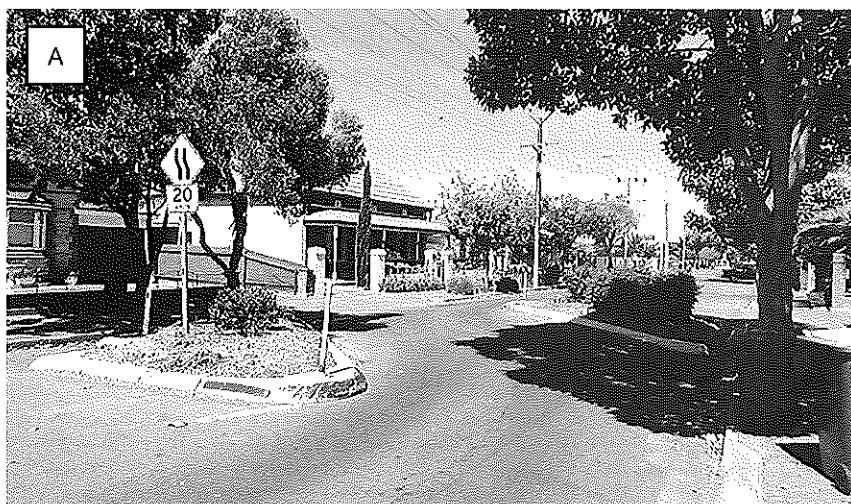
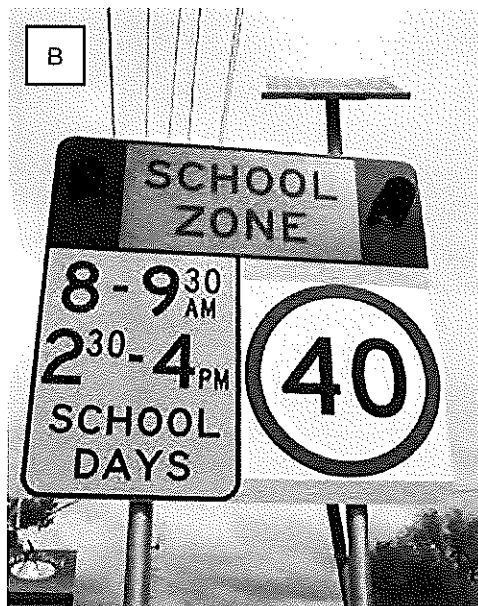
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1.5.3.7 Identify the safety devices in the photos and explain how each increase the safety of both drivers and pedestrians.



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1.6 Energy consevation, force, work, kinetic and potential energy.

1.6.1 The relationship between work, energy and force.

1.6.1.1 Give a definition of work.

1.6.1.2 In general, when is work done?

1.6.1.3 A ball is thrown up into the air. Explain who or what does work on the ball during its flight.

1.6.1.4 If an object changes its position close to the surface of the Earth, the work done on it can be calculated using the formula $W = mg\Delta h$, where Δh is the change in the vertical displacement of the object. Explain why this formula cannot be used to calculate the work required to boost a satellite into a higher altitude orbit.

1.6.1.5 A spring is stretched and then let go.

(a) What does the work when it is stretched?

(b) What does the work when it returns to its original length?

1.6.1.7 Describe a situation where work is done by an applied force. Make sure you identify the force.

1.6.1.8 Describe a situation where work is done without the application of a force. Identify what does the work.

1.6.1.9 What work is done by a 15 N force which pushed a toy car 6.0 m to the west?

1.6.1.10 A car moves at a constant speed of 5.0 m s^{-1} along a 6.0 km road against frictional forces of 150 N per metre. How much work has to be done by the car to maintain its constant speed?

1.6.1.11 A car of mass 800 g, moving at 2.0 m s^{-1} is brought to rest by a constant force acting over a distance of 12 m.

(a) Calculate the work done by this force.

(b) Calculate the value of the applied force.

1.6.1.12 A girl of mass 45 kg is on a merry-go-round at the show. The merry-go-round moves at a constant speed of 3.0 m s^{-1} in a horizontal circle of radius 15 m.

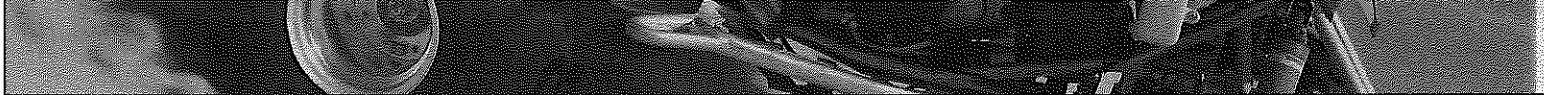
- (a) What is the net force acting on the girl?
- (b) How much work does this force do in half a revolution?
- (c) How much work is done on the girl by this force each complete revolution?
- (d) Explain your answers to (b) and (c).

1.6.1.13 Imagine identical cars travelling at 50, 60, 80 and 100 km h^{-1} .

- (a) Without doing any calculations, predict the ratio of their kinetic energies. Justify your answer.
.....
.....
- (b) Again, without doing any calculations, determine the ratio of the work the brakes need to do to stop the cars. Justify your answer.
.....
.....
- (c) Now imagine that the cars all crash into a telegraph pole and stop in 0.5 s. Calculate the ratio of the forces applied to the cars by the pole. Justify your answer.
.....
.....
- (d) Calculate how the magnitudes of the forces would be affected if the collision time was 0.1 s.
.....
- (e) Calculate the ratio of their stopping distances if the same forces were applied as braking forces. Justify your answer.
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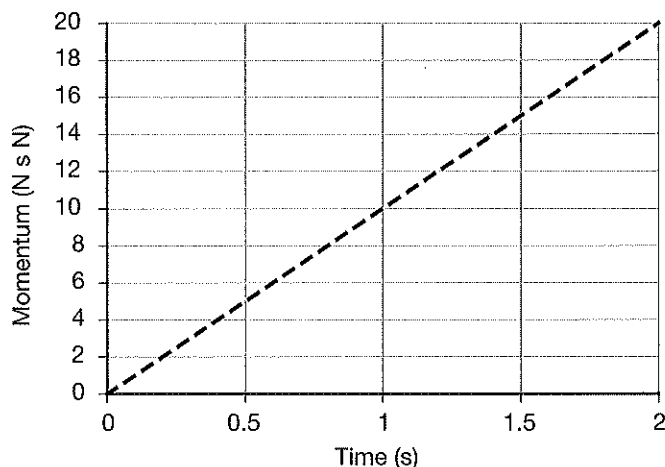
1.6.1.14 A slope rises 1.0 m for every 15 m of its inclined length. A car of mass 800 kg travels at a constant speed of 2.0 m s^{-1} a distance of 30 m up the slope against a frictional force of 200 N m^{-1} .

- (a) Calculate the work done in overcoming friction.
- (b) Calculate the total work done in moving the car 30 m up the incline.



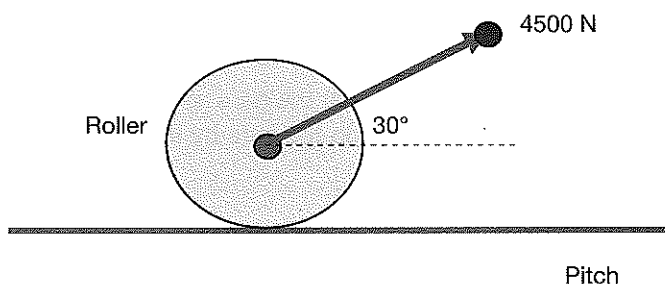
1.6.1.15 The graph shows how the momentum of a 0.6 kg object changed over time. Calculate the:

- (a) Impulse applied to the object.
.....
- (b) Final velocity of the object.
.....
- (c) Average force applied to the object.
.....
- (d) Object's average acceleration.
.....
- (e) Total distance the object travelled.
.....



- (f) Object's final displacement.
- (g) Work done by the force.

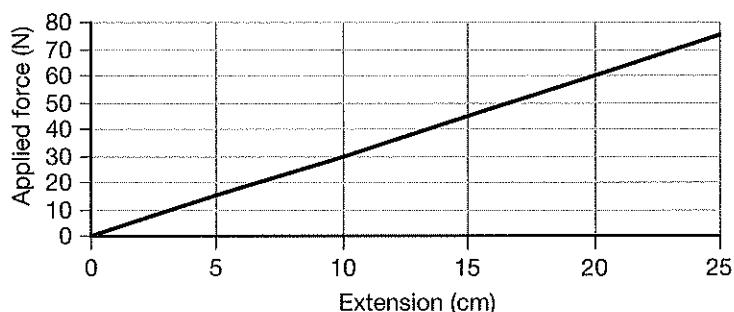
1.6.1.16 A roller is used on a cricket pitch between innings. It is moved by a constant force of 4500 N at 30° to the horizontal as shown in the diagram. The roller moves a distance of 28 m each time it rolls up or down the pitch. Calculate the work done in moving the roller from one end of the pitch to the other.



1.6.2 Force-displacement graphs.

1.6.2.1 The graph shows how a spring extends when a force is applied to it.

- (a) How much work has to be done to stretch the spring by 20 cm?
.....
- (b) What would be the elastic potential energy stored in the spring when the extension was 14 cm?
.....

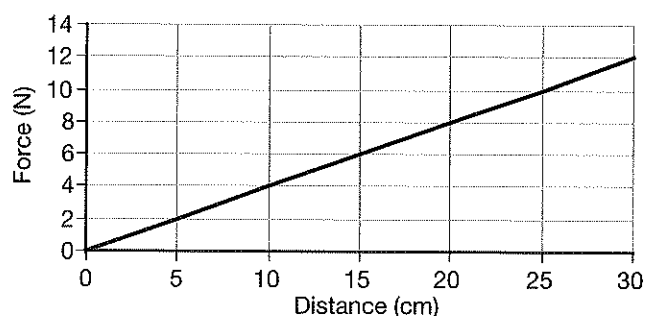


- (c) How much work would need to be done to stretch this spring by 1.0 m? (Note: This value is known as the spring constant of the spring.)
.....
- (d) How much work would be done in stretching the spring from 15 cm to 25 cm?
.....

1.6.2.2 The graph shows the force required to lift a toy car various distances above the benchtop.

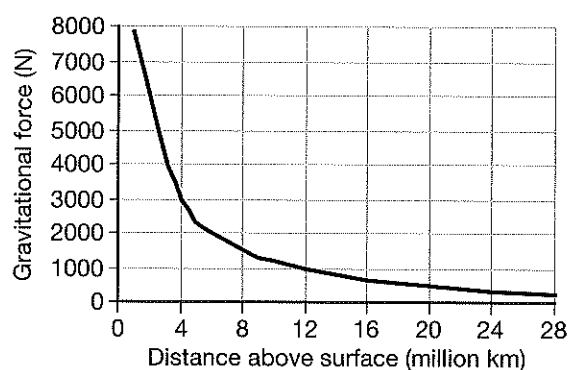
- (a) How much gravitational potential energy does the car have when it is 25 cm above the bench?

- (b) Calculate the mass of the toy car.



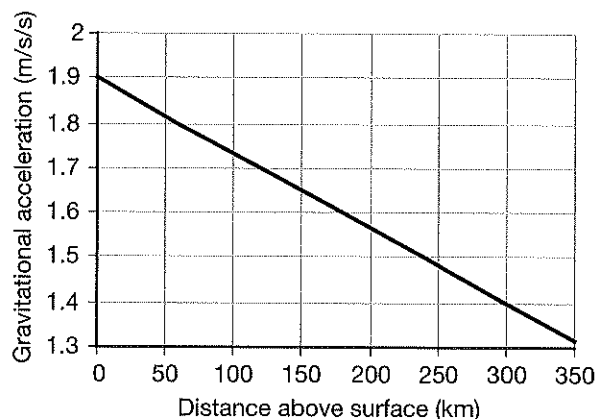
1.6.2.3 The graph shows how the strength of the gravitational force acting on an 800 kg space probe varies with distance above the surface of the Earth.

If the maximum height of the probe is 2.8×10^6 km, estimate its gravitational potential energy relative to the surface of Earth.



1.6.2.4 The graph shows how the gravitational field on a 25 kg probe fired vertically upwards from the surface of a small moon orbiting planet X varies with distance from the surface. The moon has no atmosphere.

- (a) Show that the area under the graph multiplied by the mass of the probe is equal to the gravitational potential energy of the probe.



- (b) Use this concept to calculate the gravitational potential energy of the probe when it is 200 km above the surface.

- (c) If the maximum height reached by the probe was 350 km, and assuming that it has no engines of its own, calculate the speed at which it was projected from the surface.



1.6.2.5 The graph shows the work done by two horses pulling a loaded cart along a horizontal road at constant velocity.

(a) How much work has been done by horse X?

.....

(b) How much work has been done by horse Y?

.....

(c) What total work has been done on the cart?
Explain your answer.

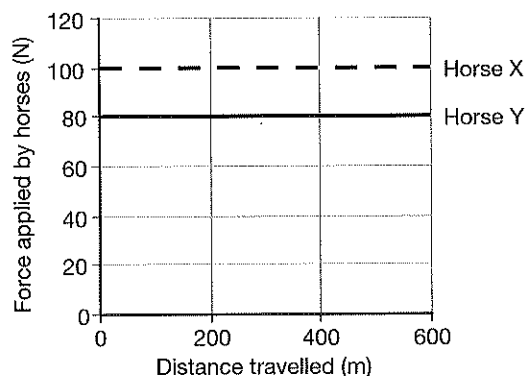
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(d) On what has work been done?

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1.6.2.6 The graph shows how the length of a spring changes as the applied load increases.

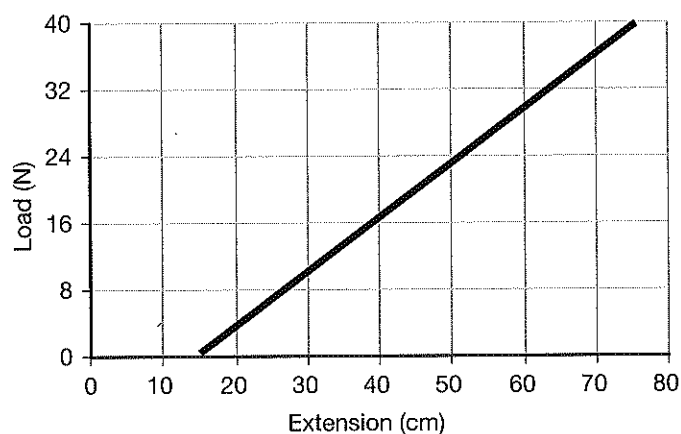
(a) Determine the value of the force needed to extend this spring by 1 m.

.....

(b) Determine the elastic potential energy stored in the spring when the load is 40 N.

.....

.....



1.6.2.7 The graph shows how the force acting on a toy car varies with distance. The car was initially at rest.

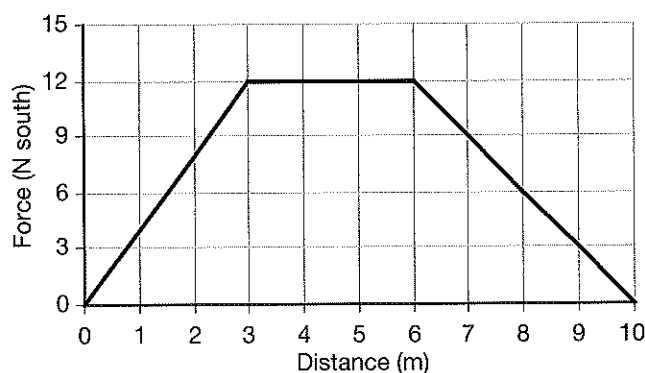
(a) How much work was done on the car in moving it the first 3.0 m?

.....

(b) What would be the kinetic energy of the car after 10 s?

.....

.....



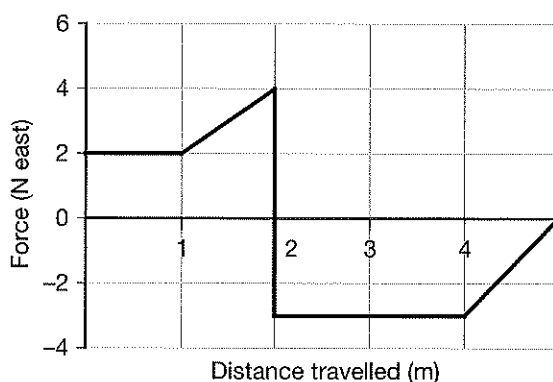
1.6.2.8 The graph shows the force applied to a toy car of mass 1.5 kg. The car was initially at rest on a smooth surface.

(a) What is the acceleration of the car at time 1.5 s?

(b) What total work was done on the car by the forces acting on it over 5 s?

(c) By how much did the kinetic energy of the car change during the 5 s?

(d) What was the final velocity of the car?



1.6.3 Power.

1.6.3.1 Define power and indicate the SI unit we use to measure it.

1.6.3.2 Two machines can do the same work in the same time. Does this mean that they develop the same power? Explain your answer.

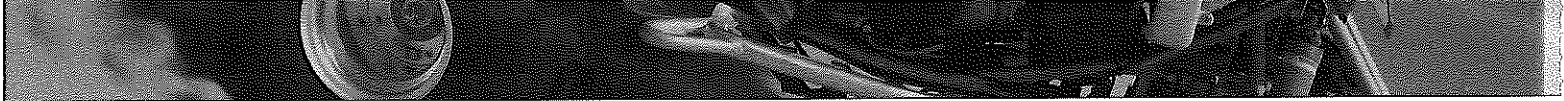
1.6.3.3 Machine X can do 400 joules of work in 2 minutes. Machine Y works at a rate of 3.5 W. Which machine has the higher power rating? Justify your answer.

1.6.3.4 An applied force does 600 J of work in 30 seconds. What power does it develop?

1.6.3.5 Explain the difference between the power developed by a machine and the work done by that machine.

1.6.3.6 A car of mass 750 kg accelerates from 10 m s^{-1} to 30 m s^{-1} in 15 seconds. What power is developed by its engine during this time?

1.6.3.7 A car hoist lifts a 1250 kg car from floor level to a height of 2.5 m in 20 s. What power is developed by the hoist?

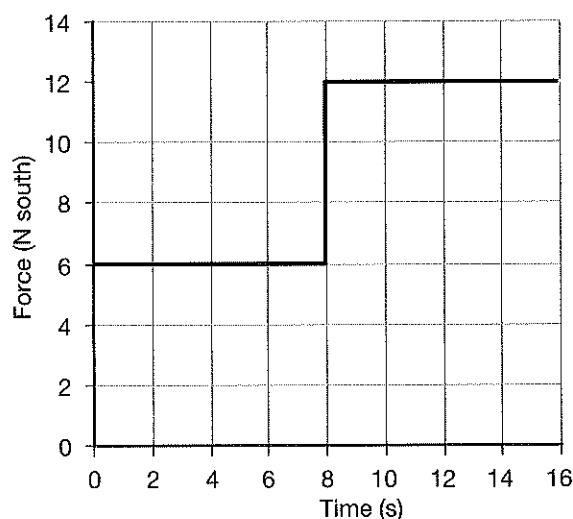


1.6.3.8 A 60 N force is applied to a toy car and this accelerates it from rest to 18 m s^{-1} .

- (a) What power is developed by the force?
- (b) How much work does the force do on the car if this takes 6 s?

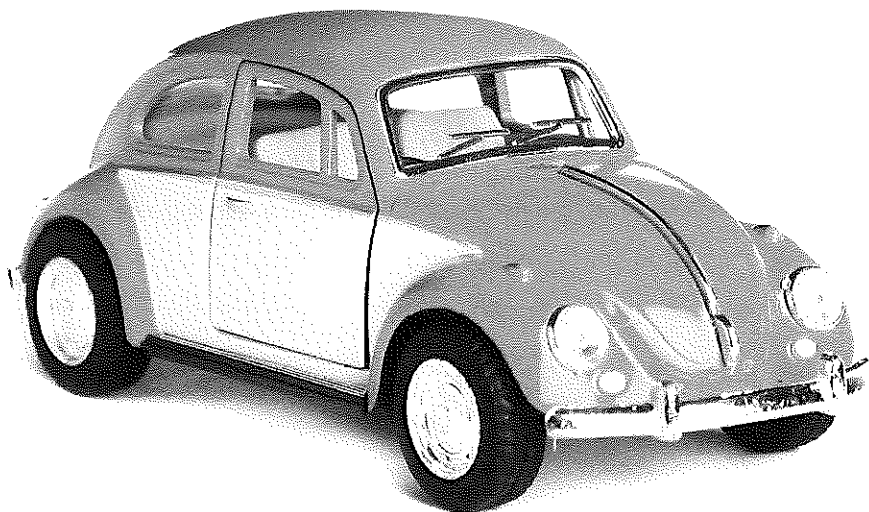
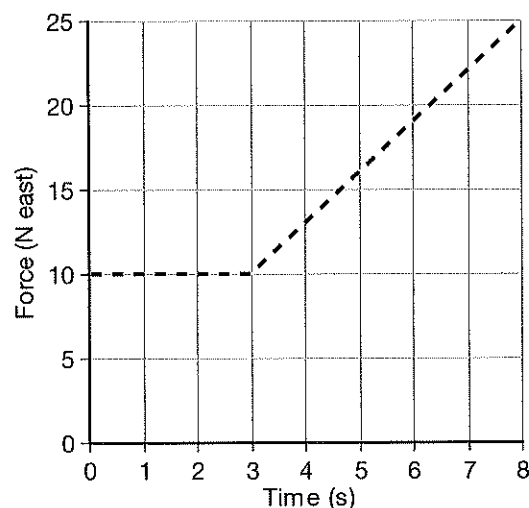
1.6.3.9 The graph shows the relationship between the force acting on a 3.0 kg toy car and the time the force acts. The car was initially at rest. Calculate:

- (a) The total impulse applied to the car.
.....
- (b) The final velocity of the object.
.....
- (c) The average power developed by the force.
.....



1.6.3.10 The graph shows the force on a mass of 2.5 kg over time. Calculate the:

- (a) Impulse applied in the first 2.5 s.
.....
- (b) Total impulse applied.
.....
- (c) Mass's change in velocity after 8 seconds.
.....
- (d) Average power developed by the force if the object was initially at rest.
.....



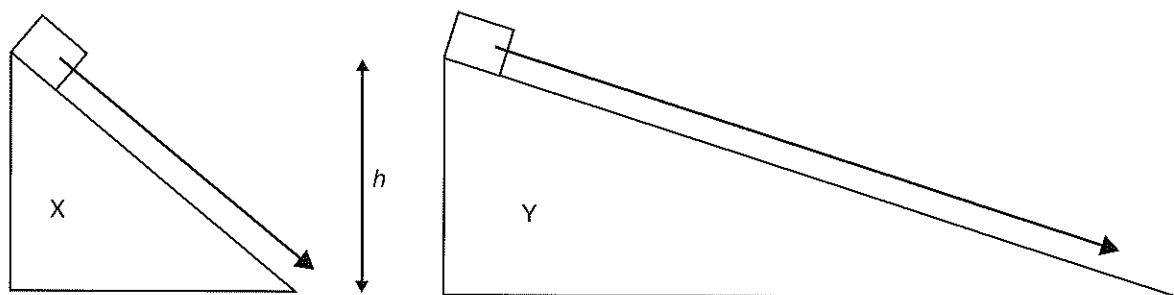
1.6.4 Energy conservation including gravitational potential, kinetic and elastic energies.

1.6.4.1 Define kinetic energy.

1.6.4.2 Define gravitational potential energy.

1.6.4.3 State the principle of conservation of energy.

1.6.4.4 Consider identical masses, mass m kg sliding to the bottom of the two frictionless inclined planes shown below. The vertical drop for each mass is the same, equal to h metres. Incline Y is longer than incline X.

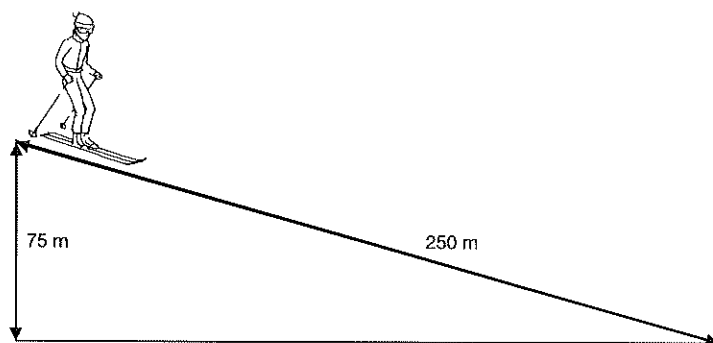


Use the law of conservation of energy to show that the speed of the blocks at the bottom of the inclines is independent of both the mass of the blocks and the distance they slide down the incline.

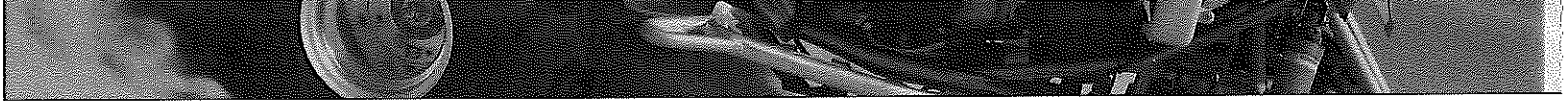
1.6.4.5 A skier of mass 55 kg at the top of a 75 m high slope skis down the 250 m incline. The snow offers a frictional force of 0.75 N.

(a) Calculate the gravitational potential energy of the skier at the top of the slope.

(b) How much work is done in overcoming friction as the skier moves down the incline?



(c) What will be the speed of the skier at the bottom of the incline?

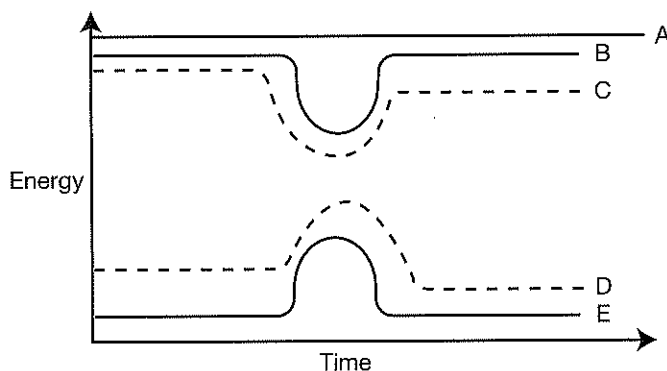


- 1.6.4.6** A 1.5 tonne pile driver falls from a distance of 5.0 m onto a pile. Each blow drives the pile 10 cm into the earth.
- (a) How much gravitational potential energy does the pile driver have at its highest position?
 - (b) What will be the speed with which the pile driver hits the pile?
 - (c) If 15% of the energy of the pile driver is lost during the collision with the pile, calculate the resistance the earth provides against the movement of the pile.
-

- 1.6.4.7** A toy rocket of mass 175 g is projected vertically into the air by a spring loaded device. The spring is compressed 12.0 cm by a force of 420 N before it is released.
- (a) Determine the maximum height the rocket will reach.
 - (b) Calculate the maximum speed of the rocket.
 - (c) Calculate the speed of the rocket when it is at an altitude of 5.0 m.

- 1.6.4.8** Consider the following graphs.
- Which of the lines on the graph represents:

- (a) The total momentum of the objects involved in an inelastic collision?
 - (b) The total momentum of the objects involved in an elastic collision?
 - (c) The total kinetic energy of the objects involved in an inelastic collision?
-



- (d) The total kinetic energy of the objects involved in an elastic collision?
- (e) The total potential energy of the objects involved in an inelastic collision?
- (f) The total potential energy of the objects involved in an elastic collision?

- 1.6.4.9** A catapult works with the aid of a spring with a spring constant of 280 N m^{-1} .
- (a) Calculate the amount of energy stored in the spring when it is compressed 40 cm.
 - (b) What would be the velocity with which a 200 g projectile leaves this catapult?

- 1.6.4.10** A 9.6 cm spring is hung vertically and a 50 g mass carrier is hung from it. This extends its length to 10.2 cm. When an additional 50 g mass is added to the mass carrier, the new length of the spring is 10.8 cm.
- (a) Predict the length of the spring when it has a total of 250 g hanging from it.
 - (b) Calculate the value of the spring constant for this spring.
 - (c) How much elastic potential energy is stored in the spring when it has been extended 12.5 cm?
-

- (d) How long would the spring be if it had 1.25 J of elastic energy stored in it?

- 1.6.4.11** A tennis ball is dropped from a height of 3.0 m, hits the surface of the court and bounces back up to a height of 2.1 m. Explain whether or not this contravenes the law of conservation of energy.

- 1.6.4.12** A toy train of mass 200 g is moving across a frictionless surface at a constant speed of 0.6 m s^{-1} . The train crashes into a spring which has a spring constant of 15 N m^{-1} . The spring is attached to a solid barrier as shown in the diagram.



- (a) Calculate the initial kinetic energy of the train.
- (b) Calculate the maximum distance the train will compress the spring.
- (c) Calculate the speed of the train at the instant the spring has been compressed 4.0 cm.

- 1.6.4.13** A 100 g glider on a linear air track is moving at 0.5 m s^{-1} towards a stationary 200 g glider. Magnets mounted on the gliders prevent them from actually colliding, but they undergo an elastic interaction. Determine the velocity of each glider after this interaction.

- 1.6.4.14** A block of wood of mass 1.5 kg is suspended from a fixed point by two light strings so that it is able to swing through the arc of a circle. A bullet of mass 25 g, travelling horizontally at 300 m s^{-1} embeds in the block. The block swings and rises through a vertical height of ' h ' before coming to rest.

- (a) Calculate the kinetic energy of the bullet before impact with the block.
- (b) Calculate the speed of the bullet and block after impact.
- (c) Calculate the value of ' h '.
- (d) Calculate the percentage of the energy 'lost' during the collision.
- (e) Where has this energy gone?

- 1.6.4.15** A girl throws a ball up into the air and across the lawn to her friend. Her friend catches it before it hits the ground. Explain the energy changes that occur in the ball in this process, what work is done and who or what does this work.

- 1.6.4.16** A basketball player bounces a 500 g ball on the court floor. Just before it hits the floor it is moving at 2.5 m s^{-1} . After it bounces it leaves the floor at 2.4 m s^{-1} .

- (a) Calculate the initial momentum of the ball.
- (b) Calculate the final momentum of the ball.



- (c) Does this contravene the law of conservation of momentum? Explain your answer.
-
- (d) Calculate the kinetic energy of the ball just before it hits the court.
- (e) Calculate its kinetic energy after hitting the court.
- (f) Does this situation contravene the law of conservation of energy? Explain.
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1.6.4.17 A bungee jumper jumps from a bridge. Explain the energy changes that occur, what work is being done and what does the work as the jumper completes his jump.

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1.6.4.18 A car veers off the roadway and crashes into a tree. Outline the energy changes that occur, what work is done and what does the work.

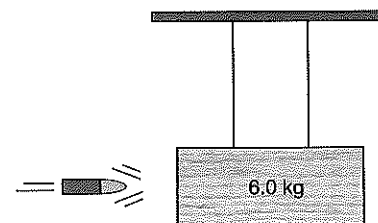
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1.6.4.19 A 6.0 kg block of wood is suspended by two light strings so that it can swing through the arc of a circle. A 50 g bullet, travelling at 310 m s^{-1} is fired at the block, hits it and embeds in it. The block swings and rises through a vertical height of h metres before coming to rest.



- (a) Calculate the initial kinetic energy of the bullet.
-
- (b) Determine the speed of the block immediately after the bullet embeds in it.
- (c) Calculate the kinetic energy of the bullet and block immediately after the bullet embeds in the block.
-
- (d) Calculate the percentage of kinetic energy lost in this collision.
- (e) Where has this energy gone?
-
- (f) Find the value of h

1.7 Elastic and inelastic collisions.

1.7.1 Elastic and inelastic collisions.

1.7.1.1 Recall the law of conservation of energy.

1.7.1.2 Describe the conditions under which the energy of an isolated system will be conserved during a collision.

1.7.1.3 Describe the conditions under which the kinetic energy of an isolated system will be conserved during a collision.

1.7.1.4 Clarify the concepts of an elastic collision and an inelastic collision.

1.7.1.5 A 1200 kg car moving at 60 km h^{-1} collides with an 800 kg parked car.

(a) As a result of this collision, the parked car is pushed forwards at 40 km h^{-1} . Determine what happens to the other car. Give your answer in km h^{-1} and m s^{-1}

(b) Determine whether the collision is elastic or inelastic.

1.7.1.6 A $6.0 \times 10^3 \text{ kg}$ train engine, moving at 2.5 m s^{-1} collides, and couples with a stationary carriage. They move at 1.2 m s^{-1} .

(a) Calculate the carriage's mass.

(b) Determine whether the collision is elastic or inelastic.

1.7.1.7 A 900 kg car, moving at 6.0 m s^{-1} east collides with a 150 kg motorbike moving west at 8.0 m s^{-1} . The car slows to 4.0 m s^{-1} .

(a) Calculate what happens to the motorbike.

(b) Determine whether the collision is elastic or inelastic.

1.7.1.8 A 45 kg go-cart crashes into a 30 kg go-cart moving in the opposite direction at 3.0 m s^{-1} . Both carts rebound at 2.0 m s^{-1} .

(a) Calculate the initial speed of the 45 kg go-cart.

(b) Determine whether the collision is elastic or inelastic.

1.7.1.9 A 350 g toy train engine is moving at 12 cm s^{-1} when it collides, and couples with a 200 g carriage moving at 5 cm s^{-1} in the same direction.

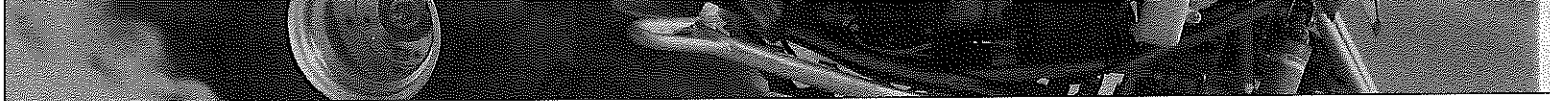
(a) Calculate their combined speed.

(b) Determine whether the collision is elastic or inelastic.

1.7.1.10 A 400 g toy engine pulls four 150 g carriages. They move at 0.3 m s^{-1} when they run into, and couple with, two identical carriages moving at 0.15 m s^{-1} in the same direction.

(a) Calculate the speed of the combination.

(b) Determine whether the collision is elastic or inelastic.



1.7.2 Analysing experimental collision data.

1.7.2.1 The diagrams show the results of an experiment where trolley X, mass 400 g, collided with trolley Y, 550 g. The results were produced using 15 Hz stroboscopic photography. Use a scale where 10 cm = 1 m.

Before collision

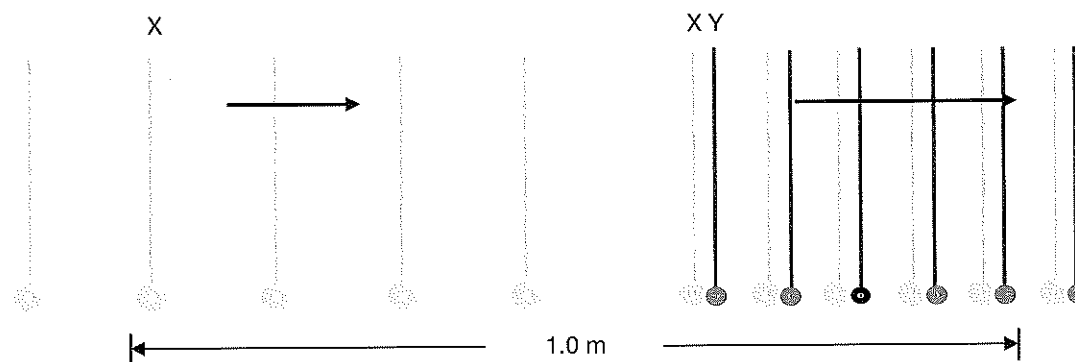


After collision



- (a) Analyse the results to determine if momentum was conserved.
-
-
- (b) Analyse the results to determine if kinetic energy was conserved, and from this indicate if the collision was elastic or inelastic.
-
-

1.7.2.2 The diagram below represents a stroboscopic photograph of a collision which used two very long pendulums, X and Y. X was swung from the right to collide with Y which was initially stationary. The mass of the bob on pendulum X was 300 g and the strobe frequency was 12 Hz.



- (a) Analyse the results to find the mass of the bob on pendulum Y.
- (b) Determine whether the collision was elastic or inelastic.
-
-
- (c) Why do you think the question refers to 'very long pendulums'?
-
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